Topping *et al. Int J Educ Technol High Educ* (2025) 22:3
https://doi.org/10.1186/s41239-024-00501-1

International Journal of Educational Technology in Higher Education

RESEARCH ARTICLE Open Access

Enhancing peer assessment with artificial intelligence

Keith J. Topping[1,2*], Ed Gehringer[3], Hassan Khosravi[4], Srilekha Gudipati[3], Kaushik Jadhav[3] and Surya Susarla[3]

*Correspondence: k.j.topping@dundee.ac.uk

[1] University of Dundee, Dundee DD1 4HN, Scotland
[2] RUDN University, Moscow, Russian Federation
[3] North Carolina State University, Raleigh, NC, USA
[4] University of Queensland, Brisbane, Australia

Abstract

This paper surveys research and practice on enhancing peer assessment with artificial intelligence. Its objectives are to give the structure of the theoretical framework underpinning the study, synopsize a scoping review of the literature that illustrates this structure, and provide a case study which further illustrates this structure. The theoretical framework has six areas: (i) Assigning Peer Assessors, (ii) Enhancing Individual Reviews, (iii) Deriving Peer Grades/Feedback, (iv) Analyzing Student Feedback, (v) Facilitating Instructor Oversight and (vi) Peer Assessment Systems. The vast majority of the 79 papers in the review found that artificial intelligence improved peer assessment. However, the focus of many papers was on diversity in grades and feedback, fuzzy logic and the analysis of feedback with a view to equalizing its quality. Relatively few papers focused on automated assignment, automated assessment, calibration, teamwork effectiveness and automated feedback and these merit further research. This picture suggests AI is making inroads into peer assessment, but there is still a considerable way to go, particularly in the under-researched areas. The paper incorporates a case study of the RIPPLE peer-assessment tool, which harnesses student wisdom, insights from the learning sciences and AI to enable time-constrained educators to immerse their students in deep and personalized learning experiences that effectively prepare them to serve as assessors. Once trained, they use a comprehensive rubric to vet learning resources submitted by other students. They thereby create pools of high-quality learning resources which can be used to recommend personalized content to students. RIPPLE engages students in a trio of intertwined activities: creation, review and personalized practice, generating many resource types. AI-driven real-time feedback is given but students are counseled to assess whether it is accurate. Affordances and challenges for researchers and practitioners were identified.

Keywords: Peer assessment, Artificial intelligence, Theory, Scoping review, Case study

Introduction

Peer assessment is an arrangement for learners to consider and specify the level, value or quality of a product or performance of other equal-status learners (Topping, 2023). This may be via grades, scores or marks, or elaborated feedback, or both. It has been shown to be effective (e.g., by Li et al., 2020a). It gives assessors a chance to deepen their understanding of the content, enhance evaluative judgment and writing skills, and learn

Springer Open

© The Author(s) 2025. **Open Access** This article is licensed under a Creative Commons Attribution 4.0 International License, which permits use, sharing, adaptation, distribution and reproduction in any medium or format, as long as you give appropriate credit to the original author(s) and the source, provide a link to the Creative Commons licence, and indicate if changes were made. The images or other third party material in this article are included in the article's Creative Commons licence, unless indicated otherwise in a credit line to the material. If material is not included in the article's Creative Commons licence and your intended use is not permitted by statutory regulation or exceeds the permitted use, you will need to obtain permission directly from the copyright holder. To view a copy of this licence, visit http://creativecommons.org/licenses/by/4.0/.

to offer constructive feedback. Assessees, on the other hand, can receive timely and personalized feedback from a diverse group of peers, fostering a more open and reciprocal exchange of viewpoints. Furthermore, instructors benefit by reducing their workload in marking (or "grading") students' work.

However, there are some common concerns associated with the use of peer assessment. Students might not possess the depth of understanding required to adequately judge their peers' work, which could influence the accuracy and fairness of evaluations. There is potential for bias and favoritism, variability in assessment standards, inconsistent feedback quality, unequal participant engagement, possible stress and anxiety, and lack of training in how to give constructive feedback. Additionally, peer assessment becomes difficult to manage in very large classes. Can any of these problems be resolved with artificial intelligence? Might AI offer scaffolding of their initial evaluation attempts? Might AI offer calibration of their initial evaluation attempts to counter favoritism or balance inconsistency? Might AI offer neutral support to counter stress and anxiety?

Artificial Intelligence (AI) is the ability of a computer (or a robot controlled by a computer) to do tasks that were usually done by humans because they (used to) require human intelligence (such as reasoning, learning and conversation) or involve data whose scale exceeds what humans are able to analyze (Topping, 2024). AI has been shown to be effective in education (e.g., by Ouyang et al., 2022) for example, in automatic assessment and making resource recommendations based on student characteristics. The rapid advancement of generative AI, exemplified by OpenAI's ChatGPT, has transformed user interaction with technology, for both good and ill. Its increasingly widespread use stems from its ability to respond to a variety of tasks. Given the right prompt, it can produce outputs for writing essays, computer code, develop step-by-step solutions to a range of problems and even compose music in different genres. Generative AI's potential has led to enquiry into how we may establish human-AI collaborations (e.g., Hwang et al., 2020; Siemens et al., 2022) to support and enhance student learning. More sophisticated techniques are being developed as we speak.

The objectives of this paper are firstly to introduce a theoretical framework about how AI might address challenges in six key procedural areas of peer assessment. Secondly, we undertake a rapid scoping review of the literature on artificial intelligence in peer assessment with reference to these six areas. Thirdly we present a case study that illustrates the application of an AI-powered peer assessment system in five of the six areas of the framework. Fourthly the paper concludes by discussing opportunities, challenges and future research needs for the effective incorporation of AI in peer assessment, as well as the implications for practitioners.

The combination of peer assessment and artificial intelligence has been explored to an extent in previous reviews. We found seven apparently relevant reviews, but most of them focused on digital technology supporting learning in peer assessment rather than AI (e.g., Fu et al., 2019; Zhang et al., 2020; Craig & Kay, 2021; Helden et al., 2023). Mendoza et al. (2020) focused more on AI in peer assessment, but AI was used mainly in the search process and the paper was mostly about rubrics and how they might be enhanced with AI. Ocampo and Panadero (2023) reviewed the characteristics and features, learning, feedback and social interaction affordances of web-based peer assessment platforms, but this was not of AI in itself. A scoping review of 27 papers on the role

of learning analytics in improving peer assessment was offered by Misiejuk and Wasson (2023), who identified three main roles: enhancing software tools, generating automated feedback and visualizations. Four main application areas were mapped: student interaction, feedback characteristics, comparison and design. The rapid scoping review in this study goes beyond this in broadly addressing all of AI in peer assessment and noting interventions rather than merely mapping the field.

A theoretical framework for AI affordances in peer assessment

The framework (Fig. 1) is based on analysis of literature and a critical examination of gaps in the current use of AI for peer assessment. A short description of each main area within the framework is presented below.

Assigning peer assessors

AI can assist by analyzing an assessor's past performance, expertise and biases to predict the reliability and quality of future contributions. Equally, by giving feedback to assessors, it can indicate areas where assessors need to focus their efforts to overcome areas of limitation. It can also facilitate the creation of balanced teams, increasing the likelihood of a more comprehensive review by matching assessee work with assessor strengths.

Enhancing individual reviews

The quality of peer reviews provides critical insights and fosters a deeper understanding for both assessors and assessees. AI's role in this process is multifaceted. By leveraging advanced algorithms and machine learning techniques, AI can guide students in developing the necessary skills to become proficient assessors. Furthermore, AI can objectively analyze the quality of each review, identifying strengths and areas for improvement. This facilitates the provision of immediate, tailored feedback to assessors, reinforcing positive practices and correcting any mis-steps. The inclusion of AI-based assessment in collaboration with students opens the door to a hybrid assessment model,

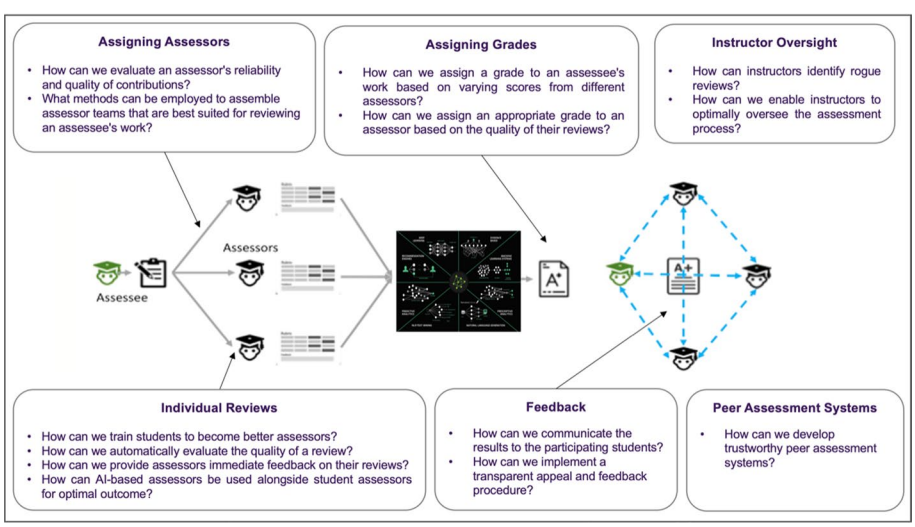

Fig. 1 A framework on AI's capacity to enhance peer assessment (building on Darvishi et al., 2022a)

directing the reviewers' attention to areas of importance, and encouraging them to provide copious feedback, blending traditional and digital assessment to enhance flexibility and accessibility—more likely to be formative, frequent and designed in collaboration with students.

Deriving peer grades/feedback

The grades that are assigned directly impact the perceived fairness and credibility of the evaluation process, and are important for maintaining motivation. However, when multiple assessors are involved, a wide range of scores can result, making the task of assigning a final grade challenging. AI's potential here lies in its ability to aggregate and analyze disparate scores and textual comments from multiple assessors in a way that is unbiased and consistent. Moreover, AI can evaluate the quality of the assessors' reviews, providing a meta-assessment (an assessment of the quality of an assessment) that not only helps in grading the assessors based on their performance but also engenders a higher standard of review.

Analyzing student feedback

Feedback needs to be not only received but also understood and acted upon by the participants. It provides the opportunity to reflect on and learn from the evaluation received. AI has the capacity to summarize and personalize the feedback in a manner that is clear and actionable. Rather than offering blanket feedback, it responds to the characteristics of the individual item to be assessed. It can also take a large amount of feedback from multiple reviewers and distill it down to a summary that can be easily digested by the author. Furthermore, it can underpin a transparent and streamlined feedback system that encourages students to engage with their assessments critically, assessing whether to act on an item of feedback.

Facilitating instructor oversight

AI can support instructor oversight of the peer assessment process by providing a dashboard of analytics, highlighting trends, flagging potential issues and suggesting areas for improvement. This can empower instructors to steer the assessment process more effectively, ensuring that it aligns with educational objectives and standards. In addition, AI can swiftly identify reviews that deviate from established norms or exhibit signs of bias, facilitating prompt intervention. This capability not only helps in maintaining the quality and fairness of assessments but also safeguards the educational value inherent in the peer review system.

Peer assessment systems

Trustworthiness is key in peer assessments, both from the perspective of those being assessed and the assessors. Through the use of sophisticated algorithms and data analysis, AI can ensure that assessment criteria are applied consistently across all evaluations, yielding feedback for both assessors and assessees regarding feedback credibility and need for improvements in feedback quality. Moreover, AI's capability to handle large datasets efficiently enables the development of more nuanced and comprehensive

assessment models. These models can take into account a wider range of factors than traditional methods.

These theoretical aspects will be further illuminated as we progress through the rest of the paper. We now turn to the rapid scoping review.

Rapid scoping review

Methodology

Google Scholar was the single database chosen for this rapid scoping review as it had been found in previous scoping reviews to generate the largest number of hits. Use of other databases would have been relatively cost-ineffective for a scoping review. It was searched over an extended period up to September 2023. The search terms "peer assessment" AND "artificial intelligence" were inserted. There are synonyms, but it was felt that for a scoping review such synonyms would not add value. Primary hits were 6,930. These were extracted up to the point where 10 consecutive pages had no relevant hits, leaving 1730 secondary hits. (Campos, et al., 2024, found that 95% of all relevant abstracts within a given dataset could be retrieved using heuristic stopping rules such as stopping the screening process after classifying 20% of records.) The titles and abstracts of these were assessed by a team. The criteria for onward inclusion were that the paper had to: refer to true AI rather than some other form of digital technology, refer to higher education, have been published in English in the last 10 years, and include quantitative and/or qualitative data supporting its conclusions. Any form of quantitative or qualitative data was acceptable providing it resulted from an attempt to evaluate the method or a thorough case study. Brief exemplars of illustrative data were not acceptable.

This led to 172 tertiary hits being selected for potential inclusion. The full text of these papers was read and 79 papers (quaternary hits) selected for final inclusion which fully met the inclusion criteria. These were coded by a single expert rater with much experience in systematic reviews and meta-analyses, which for a scoping review was thought to be acceptable. Coding was into the six areas of the theoretical framework: Assigning Peer Assessors (4 papers), Enhancing Individual Reviews (7), Deriving Peer Grades/Feedback (35), Analyzing Student Feedback (19), Facilitating Instructor Oversight (4) and Peer Assessment Systems (10). Within two of these areas, further sub-categories stemming inductively from the category content were coded by the same single expert rater, into: Deriving Peer Grades/Feedback (Automated Assessment 4, Diversity of Grades and Feedback 7, Calibration 5, Fuzzy Logic and Decision-Making 8, Teamwork Effectiveness 4, MOOCs 4, Strategic and Rogue Reporting 3) and Analyzing Student Feedback (Analysis of Feedback 14, Automated Feedback 3 and Adaptive Comparative Judgment 2).

Results

Assigning peer assessors

Instructors want to assign assessors who are competent to assess the work in question. A crude way of handling this is to randomly assign more than one assessor to each piece of work, but completing multiple peer assessments increases student workload and leaves the question of how to weight different assessments of the same piece of work. Only four papers (5%) focused on assigning peer assessors. Masaki et al. (2019) proposed an

approach based on item response theory and integer programming for assigning peer assessors, but found it no more effective than random allocation and recommended the use of additional assessors from outside the group. A system of post hoc collusion detection was trialed by Wang et al. (2019a). Only two papers looked at a comprehensive system of intelligent assignment of assessors to assesses. Anaya et al. (2019) had a system of assigning peer assessors according to social networks, which was more effective than random assignment. Having divided students into four ability groups, Zong and Schunn (2023) found matching by similar ability the most effective and more effective than random allocation, except for low-ability students.

Enhancing individual reviews

Five papers (6%) focused on improving the nature of elaborated feedback from peers and two (3%) focused on problem detection (7 in total, 9%). Of the first five, four explored the nature of different aspects of student reviews, while two sought to completely automate the analysis of student elaborated feedback. Saarinen et al. (2019) developed a tool for students to generate questions, finding its output similar to that from teachers, but this was difficult to replicate across subjects. A tool for detecting arguments in elaborated feedback was designed by Hua et al. (2019), termed "argument mining". Zingle et al. (2019) used machine learning to detect suggestions in student feedback, finding that a method involving neural networks was more effective than rule-based natural language processing methods. This would allow draft reviews to be analyzed for the presence of suggestions before the review was delivered. An algorithm for automating assessment of elaborated student feedback was produced by Wei et al. (2020). Anonymizing data is an important problem, but tangential to using AI, since if only AI reads a review, it is not necessary to anonymize it. Similarly, Castro et al. (2023) designed software which automatically classified feedback comments into categories, such as "suggestions for improvement" and "interpersonal factors". The authors studied several algorithms for making this classification, finding the results as good as teacher analysis. However, such classification would not help to decide which reviews were credible.

Two papers focused on problem detection. Xiao et al. (2020a) used machine-learning text classification methods and developed an algorithm based on natural language processing for the detection of problem statements in review comments, which transferred effectively from computing classes to a writing class. The nature of helpfulness was analyzed, which could vary from student to student. The authors attempted such a task on a few models including the logistic regression classifier, random forest classifier, multinomial naive Bayes classifier and support vector machine, using a dataset from the Expertiza system. A new algorithm was proposed by Xiao et al. (2020b) which provided accurate automated grading, weak point identification on peer reviews and also identified weak points of the reviewer. The algorithm was evaluated on 200 students generating 1,714 scores. The authors found that degradation depended on how distant the other subject was from the original one.

Deriving peer grades/feedback

This was the largest area of the six, containing 35 (44%) of the papers. These were further inductively coded into sub-categories: Automated Assessment (4 papers), Diversity of

Grades and Feedback (7 papers), Calibration (5 papers), Fuzzy Logic and Decision-Making (8), Teamwork Effectiveness (4), MOOCs (4) and Strategic and Rogue Reporting (3).

Automated Assessment The topic of Automated Assessment had four papers (5%). Kalella (2009) described an automatic grading algorithm (basically a linear-equation reputation system) which performed well even though the students were on different levels. The algorithm calculated credibility factors for each student based on the grades given by other students, then corrected the given grades by putting more weight on students with high credibility value. Grades created manually were compared to ones calculated with the algorithm and correlated well. As noted by Wang (2019b), numerical rating could be influenced by raters' biased understanding of the scale based on their cultural background. Supplementing a numerical peer evaluation system with peer-to-peer comments could reduce rater bias. Wang et al. proposed a natural language processing model that (1) processed peer-to-peer comments about teammates' teamwork behaviors; and (2) converted comments into numbers that allowed for computation. Results were evaluated in CATME data and validated the system.

An approach to automatically score student responses and evaluate the accuracy of peer assessment results was proposed by Hoang et al. (2022), via a vector space model and semantic extraction for processing and evaluating students' responses. When students realized their peer assessment responses were taken into account by the system, they had more positive attitudes towards the activities. The average of five peer assessment scores per student showed significant positive changes in the experimental group compared to the control group and was closely aligned with human assessments. Hernández-González and Herrera (2023) used a Bayesian Probabilistic Graphical Model (PGM) to estimate a grade for each test and model the peer-assessment process (accounting for the degree of precision and bias of grading peers) and a selection technique to guide instructors in deciding which assignments to evaluate manually in the event of uncertainty regarding the AI grade. The instructor could decide how many tests to grade based on their own criteria of tolerable uncertainty. The authors validated the method using both synthetically generated and real data.

Diversity of Grades and Feedback This sub-category had seven papers (9%). Babik et al. (2020) noted that little research had been done on how different network structures impacted a system's ability to accurately assess the quality of the evaluated artifacts. Two network characteristics—"dispersion" and "reciprocity" – could indeed affect assessment fidelity. The authors used BayesRank—a Bayesian approach to ranked peer grading. The peer review network structure dictated which reviewers evaluated which artifacts. Each assessment depended on both the artifact's quality and the reviewer's competence. Rashid et al. (2021) analyzed rubrics to see what features enhanced the writing of quality reviews. Measurement integrity as a desideratum for peer prediction was considered by Burrell and Schoenbeck (2021). (Peer prediction, or information elicitation without verification, is a paradigm for designing mechanisms that elicit reports about questions or tasks in settings where ground truth—and therefore the possibility of spot-checking—need not exist). They performed computational experiments, both with an agent-based model and with real data, to empirically evaluate peer prediction mechanisms, and found that peer prediction mechanisms largely failed to demonstrate significant measurement integrity. The best-performing mechanisms in terms of measurement integrity

were however highly susceptible to strategic reporting (i.e., reviewing with aims other than assessment in mind or gaming the system to improve one's grade at the expense of classmates).

Reliability (i.e., consistency in ratings across peers) and validity (i.e., consistency of peer ratings with instructors or experts) were noted as key in peer assessment by Xiong et al. (2023). The authors investigated what variables influenced prediction of these factors using a large-scale cross-context dataset. While the average number of raters per assessed item consistently showed negative effects on reliability, its effect on validity was positive and direct. A larger number of peer ratings produced a more unbiased average evaluation and therefore higher validity. The number of assignments assessed was positively related to validity, but not reliability.

Other papers dealt with individual issues. Grading/feedback inconsistency in peer assessment for open-ended activities was addressed by Rico-Juan et al. (2019) using machine learning. Two different approaches with different algorithms were tested. Experiments carried out with four groups of students and two types of activities showed that the proposed approach could yield reliable results. Bawabe et al. (2021) developed UX Factor, a peer grading platform that aimed to characterize the behavior of peer reviewers when comparing two products at a time and the consistency of ranking models used to aggregate these reviews. In a course of 133 students across three design projects the system was found to be effective in eliciting high-quality feedback. Problem-based learning in groups was investigated by Babo et al. (2021), particularly the individual contribution of each student to group performance. This paper presented seven different tools, their features and functionalities. To evaluate and compare these tools, some parameters were presented and described, based on usability and user experience definitions.

Calibration Five papers focused on calibration. In the context of peer assessment, calibration is a process by which students assess exemplar artifacts before assessing the student work that they have been assigned to review. Some systems use a student's calibration score to weight that student's reviews in calculating peer grades. Other systems require a student to continue assessing exemplar artifacts until they achieve a "passing" calibration score. The first such paper describes Calibrated Peer Review, which stemmed from the late 1990s (Russell, 2013) but was focused on assessing student work in particular subjects, such as chemistry or biology. In an early paper, Hamer et al. (2005) devised a *reputation system* for automatic grade calibration in peer assessment. It addressed the issue of grade reliability and presented a novel technique for diminishing the impact of "rogue reviews." Simulations suggested the method was effective under a wide range of conditions. Cho and Schunn (2007) described SWoRD (Scaffolded Writing and Rewriting in the Discipline), a web-based reciprocal peer review system. SWoRD included algorithms that computed individual reviewers' review accuracy and was effective in improving writing quality. A reputation system which operated similarly was proposed by Lauw et al. (2007). Neither of these authors appeared aware of Hamer's et al. (2005) contribution. Knight et al. (2019) used SPARKPlus and examined data from a benchmarking task in peer assessment (involving 500 students per semester for four semesters). Students who completed the benchmarking performed better, and there was a relationship between benchmarking performance and self-assessment performance.

However, calibration does not seem a popular area for current research, although it is clearly important in reviewer training.

Fuzzy Logic and Decision-Making Fuzzy logic is a mathematical framework that is often used in conjunction with AI and is an approach to variable processing based on "degrees of truth" rather than the usual "true or false" (1 or 0) logic in the modern computer. It allows for multiple possible truth values to be processed in relation to the same variable, particularly suitable for language. Fuzzy logic attempts to solve problems with an open, imprecise spectrum of data and heuristics that makes it possible to obtain an array of accurate conclusions. There were eight papers (10%) in this sub-category, although there seemed to be no recent papers on fuzzy logic.

Four papers were concerned with the imprecision of words. Thus, Chai and Tay (2014) used perceptual computing (Per-C). Linguistic terms provided by assessors to evaluate criteria like confidence, clarity, etc. were converted into interval type-2 fuzzy sets. The footprint of uncertainty represented variability in meanings of the linguistic terms as assessed by different experts. A codebook mapped terms to their fuzzy sets. Type reduction de-fuzzified the aggregated fuzzy output into a crisp score using the Karnik–Mendel method. Case studies were reported. Later, Chai et al. (2015) proposed a new fuzzy peer assessment methodology; a synthesis of perceptual computing (Per-C) and a fuzzy ranking algorithm which considered the vagueness and imprecision of words. Performance indices reflected a student's contribution in a group and subsequently ranked the student accordingly. A case study was reported. An algorithm to calculate the viva assessment scores using Per-C was presented by Madan et al. (2015). Accuracy of answers, clarity of concepts, confidence, spontaneity, linguistic proficiency, appearance, etiquettes and pressure handling were the issues addressed. The fuzzy system was tested to evaluate the viva (oral exam) assessment scores for nine students. Rao et al. (2017) proposed a student performance evaluation method using a Fuzzy Inference System (FIS) for Network Analysis (NA). The scoring rubric aligned the rubric criteria with Bloom's Taxonomy levels. The five inputs identify, understand, apply, analyze and design/create were fuzzified using the Mamdani Fuzzy Inference System. Results were given for an experimental group, but there was no control group or comparison with other models.

Four further papers were concerned with fuzzy grades. Thus, Ngu et al. (1995) offered a very early paper about an online peer review system called 'Peers' which let graders set the weights for different evaluation criteria and then used fuzzy decision theory to finalize the grading weights for each of the aspects. Each student peer assessed two other students and assignments were also marked by teachers. All students and teaching staff were involved in determining the criteria to be used in marking, and the weighting to be attached to each criterion. The system automatically computed the final criteria weightings using fuzzy decision theory. The paper also provided data on the reliability of peer assessment and outlined student and staff reaction to its introduction. Capuano et al. (2017) proposed a new model for ordinal peer assessment based on the principles of fuzzy group decision making. Each student was asked to rank a few random submissions from the best to the worst and to specify, with a set of intuitive labels, to what extent each submission was better than the following one in the ranking. Rankings were then transformed in fuzzy preference relations, expanded to estimate missing values and aggregated. The aggregated relation was then used to generate a global ranking between

the submissions and estimate their absolute grades. Experimental results showed better performance compared with other peer assessment techniques. A fuzzy-based approach that aimed to enhance validity and reliability was proposed by El Alaoui et al. (2018). The authors gave illustrative examples. The application of a fuzzy-number-based questionnaire was introduced by Jónás et al. (2018) in order to enhance the reliability of peer evaluations. The membership function of the fuzzy number was composed of an increasing and decreasing sigmoid membership function associated with Dombi's intersection operator. This allowed peer reviewers to express their uncertainty and the variability of the reviewed person's performance in a quantitative way. A case study was offered.

Teamwork Effectiveness Four papers (5%) focused on AI for identifying and managing team effectiveness. Petkovic et al. (2012) defined teamwork practices as the abilities: (i) to learn and effectively apply software engineering processes in a teamwork setting, and (ii) to work as a team to develop satisfactory software products. The authors presented an approach based on: (a) extracting only objective and quantitative student team activity data during their project; (b) pairing these data from related independent observations and grading of student team effectiveness; and (c) applying a machine learning approach (random forest classification) to create models, ranked factors and rules that could both explain (e.g., assess) as well as predict student teamwork effectiveness. Three universities in two countries participated in the evaluation.

The application of the online tool SPARKPlus (Self and Peer Assessment Resource Kit https://sparkplus.com.au) was reported by Wu et al. (2010), together with a set of assessment criteria for group-based ($n=4$) research projects. Thirty-seven groups participated. Some groups tried to manipulate SPARKPlus so every team member got the same score, as they thought it was a summative assessment device. Shishavan and Jalili (2020) also used SPARKPlus with groups. Peer assessment scores were individualized proportionally to the average group peer assessment score given by the teacher. Results from pre- and post-assessment were compared and students' appreciation of the teamwork aspects of the courses was improved. Automatic assessment of teamwork was investigated by Ahuja et al. (2020a), who attempted to determine (a) whether there were any characteristics common to the online discussion behaviors displayed by high-performing vs non-high-performing student team members, and (b) whether high-performing vs non-high-performing student team members shared any teamwork-exogenous attributes. The authors found that the features of team member communication that best predicted team member performance were sentence length and the number of words that contributed to the team's discussion.

MOOCs Four papers (5%) were wholly concerned with MOOCs, which are such a special context that the papers were grouped together. Kulkarni et al. (2013) reported on large online classes where peer assessment showed high correspondence with staff assessments. Three experiments were performed to improve grading accuracy. Giving students feedback about their grading biases increased accuracy. Short feedback snippets provided more qualitative peer feedback. A data-driven approach was introduced that highlighted high-variance items for improvement. Rubrics that used a parallel sentence structure, unambiguous wording and well-specified dimensions had lower variance. Algorithms for estimating and correcting for grader biases and reliabilities were developed by Piech et al. (2013), showing significant improvement in peer grading

accuracy on real data with 63,199 peer grades. They related grader biases and reliabilities to other student factors such as student engagement and performance as well as commenting style. An AI system for a MOOC-for-credit course to address both scale and endorsement was implemented by Joyner (2018). Students in the online course achieved comparable learning outcomes, reported a more positive student experience and identified AI-equipped programming problems as the primary contributor to their experiences. Sciarrone and Temperini (2020) presented a web-based system that simulated a MOOC class. This allowed teachers to experiment with different pedagogic strategies based on peer assessment. The teacher could observe the dynamics of the simulated MOOC, based on a modified version of the K-NN algorithm. A first trial of the system produced promising results.

Strategic and Rogue Reporting This sub-category had three papers (4%), two of which had similarities. Kumar et al. (2020) used the Moodle workshop tool for anonymous peer assessment and comparison with teacher assessment across 14 campuses. Moodle workshops are a very basic peer-assessment tool, providing nothing more than a way for students and instructors to view the reviews done by student peers, and so are not AI assisted. Eighty percent reliability was found. Strategic behavior in peer assessment was considered by Stelmakh et al. (2021), especially when peers were in competition with each other. In the latter case, assessors might be incentivized to misreport evaluations in order to improve their own final standing. The authors offered a statistical model for identifying such outliers which had strong detection power. Similarly, Badea and Popescu (2022) used a hybrid approach to mitigate rogue review behavior, whether due to laziness, malevolence, retaliation or dishonesty. A mechanism was proposed combining automatic labeling of suspicious reviews with manual analysis of content by the teacher. In addition, dedicated prompts were displayed to the students, providing specific recommendations for revising potentially rogue reviews. The results of a pilot study were reported.

Analyzing student feedback

This area had 19 papers (24%). Most of these concerned the Analysis of Feedback (14 papers), generally post hoc but sometimes with a view to predicting future feedback. Considerably smaller sub-categories were concerned with Automated Feedback (3 papers) and Adaptive Comparative Judgment (2 papers).

Analysis of Feedback Fourteen papers (18%) reported on the analysis of student feedback. Four focused on review accuracy. Nakayama et al. (2020) discussed the best number of peers to give evaluations to each other, related to the student's proficiency and assessment capability. The number of peers assigned to the same evaluation job was controlled from three to 50 in six steps using a 10-point scale. All parameters of the models gradually decreased with the number of peers. A multi-dimensional quality control algorithm for peer assessments and text information was developed by Li et al. (2020b). The user's behavior, comment text information and other elements were combined together. The frame was a log-linear model leading to a gradient descent algorithm. When compared with traditional algorithms, the model performed better. Badea and Popescu (2020) described LearnEval and applied it to project-based learning scenarios. Each student was modeled on competence, involvement and assessment abilities. A scores

module involving visualization was incorporated. However, evaluation was only through student perceptions. Three different systems for analyzing peer comments were used by Huang et al. (2023), classifying peer comments in terms of cognitive content and affective state. The Bidirectional Encoder Representations from Transformers (BERT) model had the best results and improved feedback with significantly reduced student burnout. Individuals who received more suggestive feedback experienced a greater reduction in emotional exhaustion. By contrast, when receiving more negative feedback and/or reinforcement without guidance, learners tended to have a worse emotional experience and demonstrated poorer self-learning behavior.

Eight papers focused on review helpfulness and problem detection. Xiong and Litman (2010) developed a system to automatically identify problem localization in peer feedback. Regular Expression Features, Domain Lexicon Features, Syntactic Features and Overlapping-window Features were all deployed. The model significantly outperformed baseline. Structural features, review unigrams and meta-data were useful in automatically predicting peer review helpfulness according to Xiong and Litman (2011). Xiong et al. (2012) then applied a combination of data mining and natural language processing to automatically detect a lack of helpful elements in peer feedback. Then a model was constructed using Weka to detect target feedback features (e.g., localization information and concrete solutions). Reviews were automatically coded for these feedback features. Helpful feedback in an affective context was positive, motivating and encouraging according to Selmi et al. (2014), while unhelpful feedback was negative, bullying and demeaning. They proposed an approach using natural language processing techniques and machine learning to predict the helpfulness of given affective feedback for a learner based on the feedback content and the learner's affective state. The Naïve-Bayes algorithm provided the best performance. According to Rashid et al. (2022), quality feedback focused on detecting problems, providing suggestions or pointing out where changes needed to be made. They found students were likely to find a review helpful if a suggestion was provided along with the identified problem. Detection of problems in reviews was also pursued by Xiao et al. (2022). They tried several machine-learning and neural network models, finding that their proposed new algorithm was as or more efficient. Saccardi et al. (2023) designed a system for identifying students' group work problems which addressed feedback on teamwork, rather than feedback on an artifact produced by an individual or a team. E-Mate was a virtual agent incorporating visualization that provided initial feedback on the assessment. Automated evaluation of review helpfulness was investigated by Liu et al. (2023), with the help of deep learning models and natural language processing techniques. The study evaluated the performance of self-training in predicting the helpfulness of peer reviews. Also, an advanced self-training framework was proposed by tailoring knowledge distillation and noise injection to further improve model performance.

Two papers concerned the quality of elaborated feedback. Patchan et al. (2016) tested an updated theoretical model including a broad set of features and considered not only students' likelihood of implementing a comment but also the quality of their revisions. They analyzed 7,500 feedback comments and each comment was coded for the presence of praise, a problem description, a suggested solution, localization, focus (i.e., low prose, high prose, substance), implementation and revision quality. Only two feedback

features increased students' likelihood of implementation (overall praise and localization), while several reduced it (mitigating praise, solutions and high-prose comments). Three conditions were then compared by Patchan et al. (2017): only rating accountability, only feedback accountability, or both rating and feedback accountability. Peer ratings and peer feedback were coded. Constructing helpful comments had a broad influence on peer assessment and consistent ratings were grounded in this commenting. It should be noted that there is a fine line between feedback analysis (in this section) and Enhancing Individual Reviews. If feedback analysis is instant, and reviewers can see it before submitting a review, it can help reviewers to improve their reviews. This might be considered a "formative" use of feedback analysis. If it is instead presented to the instructor as a way of assessing reviewer effectiveness, then it is being used "summatively" to analyze the feedback.

Automated Feedback Disappointingly few papers (3–4%) focused on truly automated feedback, and none were of recent origin. A system for evaluating and reporting the presence of solutions in feedback at the sentence level was described by Nguyen et al. (2016). Better feedback was triggered with higher precision leading to better solutions. Using data from Expertiza and Scaffolded Writing and Rewriting in the Discipline (SWoRD) (two collaborative web-based learning applications), meta-reviews or reviews of reviews were discussed by Ramachandran et al. (2017). They evaluated feedback quality which triggered further development of feedback. The model yielded gains from baseline. Similarly, Nguyen et al. (2017) developed a system that automatically provided formative feedback on students' feedback comments, which enhanced the quality of feedback and made it more specific.

Adaptive Comparative Judgment (ACJ) There were two papers in this sub-category (3%). Demonacos and Barber (2019) explored ACJ as a medium for peer assessment. ACJ is a marking protocol in which the assessor compares two artifacts and chooses a "winner". Repeated judgements and a sorting algorithm allow marks to be derived from the rank order of scripts. Feedback can be added to each script. In this case study each student gave feedback to ten others. Students were inconsistent with one another and with staff assessment. A hierarchical marking scheme and explicit feedback guidelines were recommended. ACJ was also trialed by Ellison (2023)—avoiding rubrics but encouraging assessment by comparing different pieces of work. However, improvements in attainment were not measured.

Facilitating instructor oversight

Four papers (5%) addressed a variety of topics. "Free riding" was defined by Chiu et al. (2020) as exploiting another's work or taking credit for work done by others without contributing to a team project. An online assessment system for individual scores (OASIS) was designed to reduce student perceptions of free riding, including: contractual commitment, team discussion, median peer assessment of each member's contribution, assessment revision opportunities, conditional teacher participation and final appeal. Students in India and Hong Kong participated. Negative findings came from larger groups, lower individual monitoring of individuals and greater competence differences between members. Students who valued team projects more than others experienced fewer free riders. A tool called PuzzleMe was described by Wang et al. (2021), which

delivered live peer testing and live peer code review. Live peer testing could improve students' code robustness by allowing them to create and share lightweight tests with peers. Live peer code review could improve code understanding by intelligently grouping students to maximize meaningful code reviews. However, the evaluation was very brief. Khosravi et al. (2021) described *learner sourcing*, via an adaptive system called RiPPLE, which had been used in more than 50 courses with over 12,000 students. The paper offered data-driven reflections and lessons learned. Dhull et al. (2022) focused on finding an assignment of evaluators to submissions that maximized evaluators' expertise subject to the constraint of strategy-proofness. Several polynomial-time algorithms for strategy-proof assignment along with assignment-quality guarantees were developed and successfully trialed.

Peer assessment systems

Readers interested in this topic (10 papers, 13%) are also advised to consult Ocampo and Panadero (2023), who review a large number of peer assessment systems including many well-known ones. However, many of these did not have AI components.

Six papers reported systems labeled with catchy acronyms. Wang et al. (2012) introduced EduPCR, in which students peer reviewed programs written by other students. Teachers assessed and gave scores to students based on their performance in writing, reviewing and revising programs and their adherence to a peer code review process. When used in two courses there were significant attainment improvements. Student perceptions were also positive. Use of the Peergrade tool was reported by Sharma and Potey (2018), in which the entire process of learners' assignment submission, peer assessment, feedback to peers, reviews and assessment reports for the teacher was completely automated. Student perceptions showed 92% of learners agreed that the feedback provided by their peers was useful. The IPAC tool was introduced by Garcia-Souto (2019), which featured: customizable assessment criteria, a range of feedback to students, easy access by students (integrating with Moodle) and feedback being anonymous. IPAC software was used by different courses. He et al. (2019) proposed a novel cognitive diagnosis model named Peer-Assessment Cognitive Diagnosis Framework (PACDF)—a probabilistic graphical model characterized not only the relationships between real scores and scores given by peer assessment, but also the relationship between examinees' skill proficiency and problem mastery. Then a sampling algorithm predicted the examinees' performance. The authors adopted a Monte Carol Markov Chain sampling algorithm to estimate the parameters of the model. Experimental results show that PACDF could quantitatively explain and analyze skill proficiencies of examinees. Ravikiran (2020) also discussed auto grading and peer assessment tools (Peer Grader and Peer Studio) to handle rogue reviews and peer review improvement using natural language processing. However, the evaluation was weak. G-PAT, a tool to support group projects, was introduced by Tiew et al. (2021). A web services approach was used so that students and instructors could access the service from anywhere and anytime. Students could customize the question as they liked. Student perceptions were very positive.

The four remaining papers reported different aspects. Thus, Hsia et al. (2016) reported on peer assessment in dance classes. An online peer-feedback system was developed to compare the effects of different modes of online peer-feedback on students' dance skills

performance, learning motivation, self-efficacy, peer review quality, peer assessment correctness and online learning behaviors. A 12-week experiment was conducted comparing videos with peer comments, videos with peer ratings and videos with peer ratings plus peer comments. The latter group provided better feedback and were most aligned with teacher scores. In the system proposed by Thamizhkanal and Kumar (2020), preprocessing and noise removal was done with the help of filtering, normalization and compression. Then internal and external segmentation was deployed. Identification of a letter in the answer script was done by convolutional neural networks. Evaluation of answers was done by a simple neural network. An experiment comparing test types was in favor of the new system.

Ahuja et al. (2020b) applied various machine learning algorithms to data from the CATME system regarding the interactions of first year college students using a messaging platform when collaborating online on a team project, seeking to predict measures of team member performance. A wide range of algorithms were used. Text messages of online team member exchanges could discriminate high performing from non-high performing team members. The analysis of peer assessment interactions and phenomena could benefit from social network analysis according to Djelil et al. (2021). They applied a graphlet-based method to a dataset collected during in-class courses integrating a peer assessment platform. Students developed a positive tendency towards adopting the peer assessment process and engaged gradually with well-balanced roles, even though initially they chose mostly to be assessed by teachers and more likely by peers they knew.

Now we present a Case Study which aligns both with the areas of the Theoretical Framework and with the areas of the Scoping Review (although it only touches upon five of the six areas).

Case study

Here we present a case study using a tool called RiPPLE (Khosravi et al., 2019) that transforms student learning into an active, personalized, and social journey. At its core, RiPPLE empowers instructors and students to collaboratively curate a wealth of high-quality learning materials. These resources then serve to tailor content recommendations to students, aligning with their individual mastery levels. Through RiPPLE, students are not just passive recipients of information but active creators, honing vital skills in content creation, critical analysis, and effective communication within a supportive community. Instructors, on the other hand, find a scalable way to enrich their courses with diverse practice materials, significantly reducing the traditional time investment required. Finally, academic departments benefit by offering customized educational experiences at minimal costs. Central to achieving these ambitious objectives, as depicted in Fig. 2, are the dynamic, interconnected activities of content creation, peer review, and personalized practice sessions—each carefully orchestrated by instructors and enhanced with AI-driven insights and real-time feedback.

To achieve these goals, as shown in Fig. 2, RIPPLE engages students in a trio of intertwined activities of creation, review and personalized practice. These activities are conducted under the guidance of instructors, complemented by AI-driven real-time feedback and personalized instructions. Each is outlined in detail below.

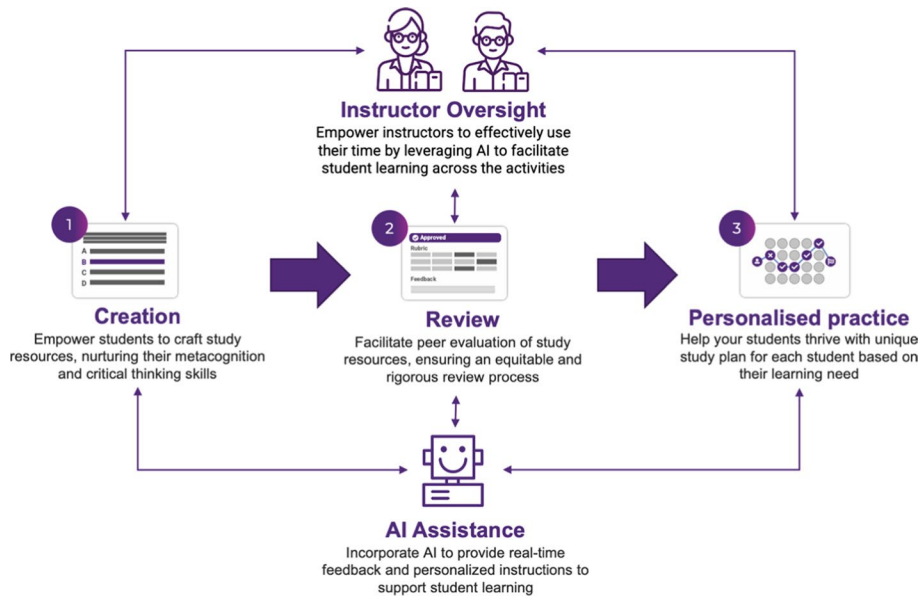

Fig. 2 Overview of the RiPPLE platform

Creation

RiPPLE actively engages students in the creation of diverse and innovative study resources, a process also known as learnersourcing (Khosravi et al., 2023). It empowers students to take an active role in their learning and fosters a deeper understanding of the subject matter as they contribute to the educational resources available to their peers. The platform supports the creation of a wide range of resource types such as multiple-choice questions, worked examples, flashcards, hotspots, as well as short and long-answer questions. This diversity allows students to engage with the content in various formats, which enhances the richness of the resources available on the platform and help cater to different learning preferences.

The concept of learnersourcing has been well-studied in the literature as an approach that enables students to produce high-quality educational materials. Previous studies, including those by Abdi et al. (2021), have shown that when students are tasked with creating learning resources, they often produce content that meets rigorous judgmental and statistical criteria, which highlights the ability of students to contribute meaningfully to their learning environment and to produce resources that are not only useful for their peers but also align with academic standards. Moreover, creating study resources has been linked to improved comprehension, and retention of content. When students engage in the process of creating study resources, they must deeply engage with the material, which leads to better internalization and understanding. This active learning process encourages critical thinking, and allows students to explore the content from different perspectives to reinforce their grasp of the subject matter.

Review

The process of peer assessment is integral to the validation and enhancement of student-created educational resources within the RiPPLE platform. Upon the creation

of resources by students, these materials are subjected to a peer review mechanism whereby peers evaluate the content's quality and relevance. The peer assessment process fosters a collaborative learning environment, where students learn to apply academic standards in a practical context, further reinforcing their learning of the course content.

Once peer assessments are completed, resources that are deemed to meet or exceed the established criteria are approved and integrated into a central repository. This repository serves as a curated collection of study materials accessible to all students, thereby enriching the overall learning experience. Conversely, resources identified as lacking in effectiveness are returned to their original authors, accompanied by constructive feedback. This feedback is instrumental in guiding the authors through the process of revising and improving their work, after which they are encouraged to resubmit their enhanced resources for another round of evaluation. By engaging students in both the creation and assessment of learning materials, they get actively involved in a dynamic educational ecosystem that emphasizes peer collaboration, continuous feedback, and the iterative refinement of knowledge.

Personalized practice

As students actively engage with approved learning resources, RiPPLE's AI algorithm continually evaluates their mastery across different course topics. This process is a critical component of the platform's adaptive learning engine, which leverages the collected data to calculate each student's knowledge level on each course topic to be able to personalize the educational experience for each student and recommend targeted learning resources that are specifically aligned with their individual learning needs and current comprehension. This approach not only helps to reinforce foundational knowledge but also facilitates more effective learning by guiding students through content that is most pertinent to their academic development at any given time (Abdi et al., 2020).

Instructor monitoring

To enhance the reliability and accuracy of the peer review process, RiPPLE incorporates an AI spot-checking algorithm. This approach plays a critical role in ensuring that the peer assessments are consistent and that the quality of the student-created resources remains high. Specifically, the AI algorithm is designed to identify resources that have been flagged as inappropriate or that exhibit a high degree of variability in peer evaluations. These are the resources most likely to benefit from further scrutiny and expert judgment. This identification process allows instructors to optimally focus their time and expertise on reviewing the resources that are most in need of their attention, rather than spending time on materials that have already been consistently rated as high-quality by peers (Darvishi et al., 2022a). This combination of AI-driven oversight and expert intervention represents a critical component of RiPPLE's strategy to balance the scalability of peer assessment while maintaining the integrity of the learning materials within the platform, ensuring that students have access to reliable, high-quality resources. Furthermore, it reinforces the educational value of the peer review process. Students are aware that their work may be subject to additional scrutiny by both AI and their instructors, which can motivate them to engage more deeply with the material and produce higher-quality resources.

AI assistance

During the creation phase, generative AI is seamlessly integrated into the platform to provide immediate feedback on the resources submitted by students. This AI-driven feedback mechanism includes a comprehensive summary that interprets the primary objective of the resource, ensuring that the content aligns with the intended learning outcomes. Additionally, the feedback highlights the resource's strengths, acknowledging well-executed elements that contribute to effective learning as well as suggestions for specific areas for improvement, offering actionable recommendations that can elevate the content's overall effectiveness and clarity. In the review phase, generative AI is once again employed to deliver real time constructive feedback, designed to identify potential areas where the review could be strengthened, such as offering more detailed analysis, providing clearer justifications, or suggesting alternative perspectives. For personalized practice, RiPPLE's AI algorithms assess students' abilities in each course topic, recommending resources that are most suitable for their current knowledge level.

RiPPLE has been adopted in over 250 subject offerings across a range of disciplines including Medicine, Pharmacy, Psychology, Education, Business, IT and Biosciences. Over 50,000 students have created over 175,000 learning resources and over 680,000 peer evaluations rating the quality of these resources. The adaptive engine of RiPPLE has been used to recommend over three million personalized resources to students. In the sections below, we discuss how RiPPLE leverages five of the six areas outlined in our proposed framework (as it is itself a Peer Assessment System).

Assigning peer assessors

In RiPPLE, the peer assessment process is carefully designed to ensure fairness and quality in evaluating student-created resources. The system operates under a double-blind protocol, meaning that both the creators and assessors of the resources remain anonymous to each other. The prioritization of resources for assessment is based on their submission time, with earlier submissions receiving precedence in the review queue. The number of assessors assigned to each resource is flexible and determined by the course instructor. They set a range specifying the minimum and maximum number of assessors required. When assessors are in agreement about the quality of a resource, a decision is finalized after reaching the minimum number of assessments. However, in instances of disagreement among assessors, the system may extend the process up to the maximum number of reviews stipulated by the instructor before reaching a decision.

A key component of this process is the use of advanced machine learning algorithms such as the graph-based trust propagation approach (Darvishi et al., 2021) to gauge the reliability of each assessor's evaluations. The system aims to include at least one assessor deemed reliable in the mix for assessing each resource. This enhances the credibility and accuracy of the peer review process, ensuring that each resource is evaluated with a balance of diverse perspectives and expert judgment. Through these mechanisms, RiPPLE not only maintains the integrity of the peer assessment process but also actively involves students in critical evaluation, fostering a deeper engagement and understanding.

Individual reviews

Individual reviews within RiPPLE are structured around a comprehensive rubric (see Fig. 3), tailored to assess various dimensions of resource quality. The rubric incorporates multiple criteria, each specifically aligned with the nature of the resource being reviewed. For example, in the case of multiple-choice questions, one criterion assesses the quality and relevance of the options provided, while in worked examples the rubric examines the clarity and comprehensiveness of the intermediate steps. These criteria are carefully designed to ensure a thorough evaluation of each resource type, addressing aspects unique to their format and purpose. After applying the criteria, students engage in a peer feedback process, which involves Identifying and articulating the positive aspects of the resource, suggesting areas of improvement and providing any additional comments that could enhance the resource's value. Upon completing the feedback, students make a judgment on the overall quality of the resource. This assessment is pivotal in determining whether a resource should be approved for use or rejected. Additionally, students are asked to rate their confidence in their assessment, providing insight into their self-evaluation skills and understanding of the review criteria.

Integration of Generative AI in Feedback RiPPLE has enhanced its review process by incorporating generative AI technology. This assists students in formulating their peer feedback. It offers immediate, constructive responses to submitted reviews, insights into areas needing improvement and strategies for enhancing the quality of peer evaluations. This feedback aims to not only enrich the learning experience but also ensure a

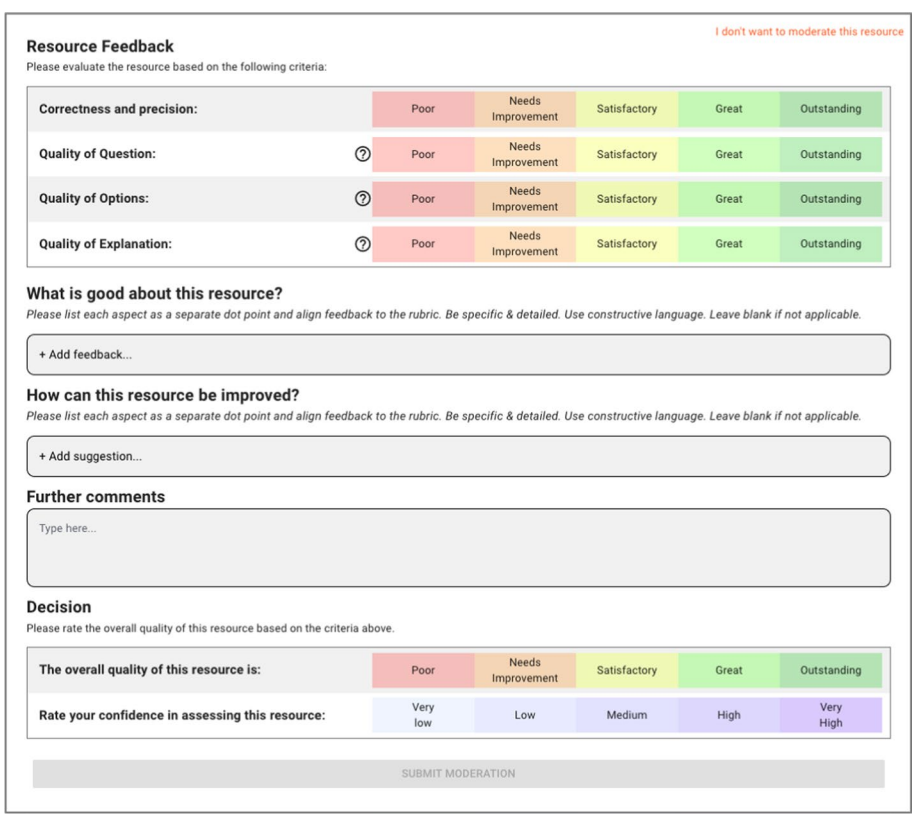

Fig. 3 The rubric and feedback form used for peer assessment in RiPPLE

higher standard of peer review by providing additional objective insights. Figure 4 shows an example of the type of feedback provided by the generative AI. The upper section of the AI feedback component emphasizes that "the feedback provided below is only a guide and may be inaccurate. Please use your domain knowledge to assess whether the feedback is correct." This statement serves as a reminder to students about the potential for inaccuracies, often referred to as 'hallucinations', in responses from large models. It underscores the necessity for students to apply their own evaluative judgment in conjunction with the AI's feedback. This section is followed by the AI introducing itself, offering suggestions for improvement. Additionally, it provides space for students to evaluate the feedback both quantitatively and qualitatively.

The AI-feedback on peer feedback feature currently has been used for one semester and has accumulated 4,553 ratings. It has achieved a commendable average rating of 4.4 out of 5, with a substantial 75% of students assigning a 4- or 5-star rating, revealing its considerable utility among the majority. However, 14% of users expressed neutrality with a 3-star rating, while 11% found it less beneficial, assigning a 1- or 2-star rating. Positive feedback, such as a student noting, "This feedback was very helpful as it gave me the answer I was looking for but didn't know how to put into words," underscores the AI's capability in assisting students to articulate their thoughts more effectively. A representative 3-star, neutral comment was: "Dot point 2 would be good to improve my feedback and make it more specific. I think the AI did not quite understand dot points 1 and 3, as

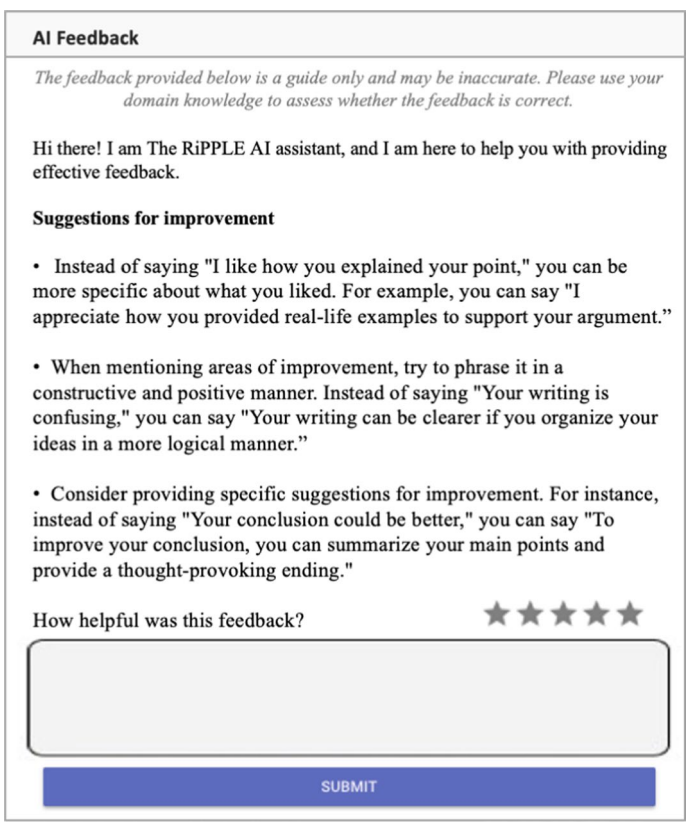

Fig. 4 The AI-feedback on peer feedback interface in RiPPLE

I feel that I clarified these in my feedback." This reflects a balanced view where students discern the utility of certain aspects of the feedback while critically evaluating others. A sample of negative feedback: "The suggestions for improvement were the same wording as my response with no explanation for why this needed improvement" highlights instances where the feedback was seen as redundant or uninformative. These varied perspectives illuminate both the strengths and areas of potential refinement for this feature.

Deriving peer grades/feedback

RiPPLE assigns each resource for evaluation to multiple reviewers (moderators), which then requires a consensus approach by optimally integrating the decisions made by multiple people towards an accurate final decision. In assessing the quality of student submissions based on scores from multiple assessors, we analyzed 18 inference models across five established categories of consensus approaches on data collected via RiPPLE (Darvishi et al., 2022a). The first utilized summary statistics like mean or median (e.g., Purchase & Hamer, 2018). The second leveraged historical performance or self-evaluation data to estimate student competence, a method often found in adaptive educational systems. The third applied probabilistic consensus techniques, assessing student reliability based on their previous reviews, a strategy effective in crowdsourcing systems (e.g., Raykar et al., 2010). The fourth employed text analysis to determine review reliability based on the comments provided, a technique successfully used in identifying reliable reviews (e.g., Kim et al., 2006). The fifth merged methods from the previous four, selected for the proven effectiveness of ensemble consensus approaches. Our findings based on data collected from 2,141 undergraduate students across five courses in creating 12,803 resources and 77,297 peer reviews revealed that review quality varied significantly, with a tendency for students to overrate submissions. Consequently, simple statistical measures like mean and median are inadequate for identifying low-quality resources. The results suggested that integrating advanced probabilistic and text analysis methods for assessing reviewer reliability and review quality enhanced performance. However, there was a clear need for continued instructor oversight and training to help students craft effective and dependable reviews (Darvishi et al., 2022a).

RiPPLE employs a graph-based trust propagation approach (Darvishi et al., 2021) to infer the reliability of each moderator's evaluation. The final decision regarding the resource is determined by calculating a weighted average of the ratings provided by peer evaluators. Figure 5 illustrates how the evaluations and inferred outcomes are communicated to both the author and the moderators. The process begins with authors and moderators being prompted to evaluate the usefulness of the feedback received from each review. They are then asked to indicate whether they agree or disagree with the inferred outcome and to provide any additional feedback. According to platform data, only 2.2% of the responses submitted by users disputed the inferred outcome (Darvishi et al., 2022b), indicating a high level of trust and accuracy in the system's decision-making process.

Instructor oversight

RiPPLE is designed with a key principle in mind, which is to empower instructors with actionable analytics and insights and to help them to maximize the efficient use of their

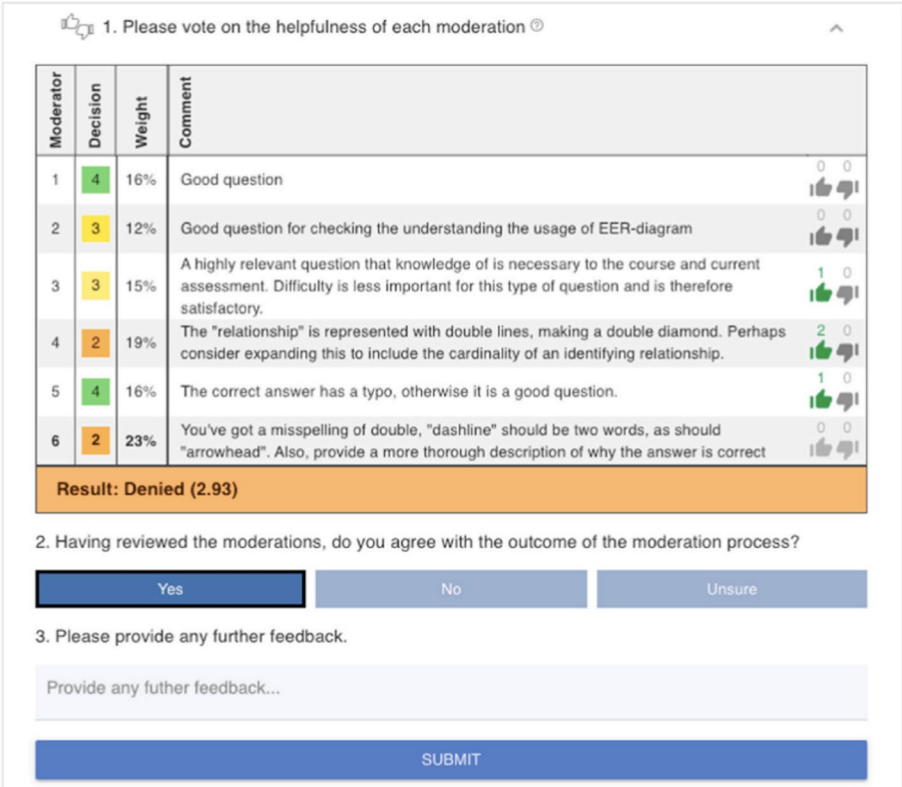

Fig. 5 RiPPLE interface displaying reviews, outcome and soliciting feedback

limited time. This is exemplified in the instructor landing page (Fig. 6). The upper section of this page presents an array of statistics, offering insights into both overall and the previous week's activities. This includes data on study sessions, and the number of resources created, evaluated, and responded to. The page then highlights the completion status of the latest assessment round, alongside weekly updates on student achievements and the most popular resources. To optimize instructor time, a section in RiPPLE is named 'Suggested Actions.' This section presents four primary types of recommendations alerting instructors to (1) inspect flagged resources that are greatly beneficial, (2) review evaluations flagged as ineffective, (3) review students who are underperforming significantly, and (4) remind students who have not completed tasks to do so. The bottom of the page includes an analytical toolbox that provides answers to a list of questions in relation to students' performance and engagement. For each of the questions, instructors can set the start and end date for data being reported to view class-level or individual-level trends using various visualization types (e.g., bar charts, box plots).

Figure 7 displays the interface used for examining resources that might be incorrect or ineffective. Broadly, it utilizes a combination of human-centric metrics (such as significant disparities in reviews or a high ratio of down-votes compared to up-votes) and data-centric metrics (like assessment items with a low discrimination index or distractors where the commonly chosen answer differs from the author's intended one). These metrics classify resources into high, medium, low or no review priority. To assist instructors in understanding why a resource has been marked for

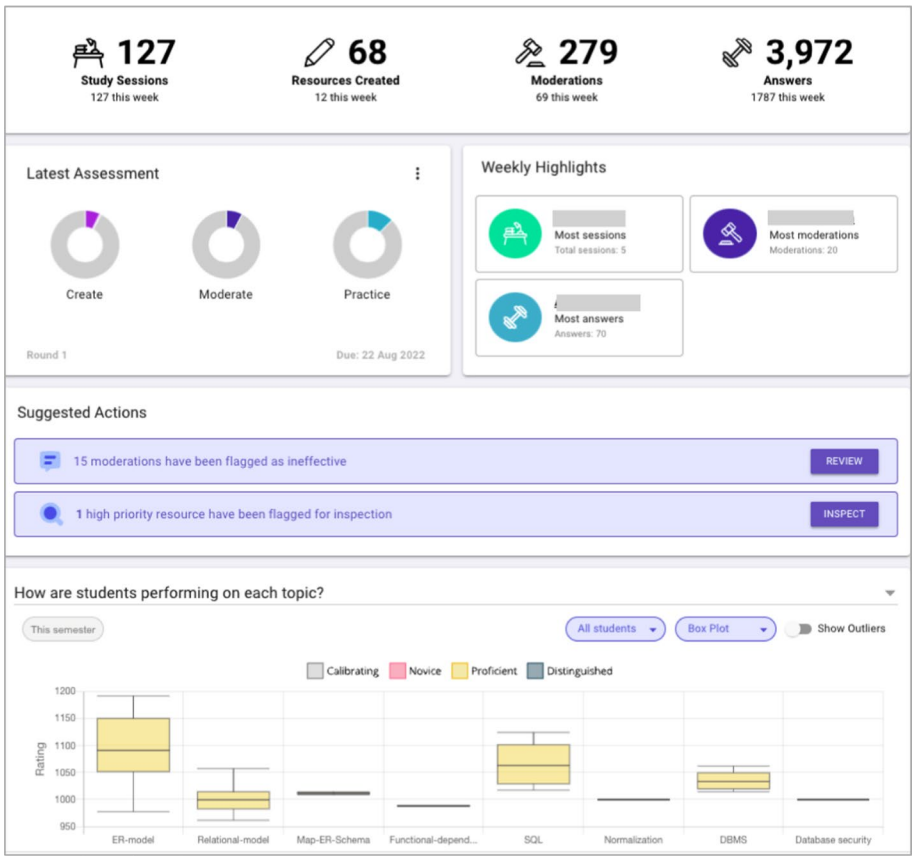

Fig. 6 Instructor landing page in RiPPLE

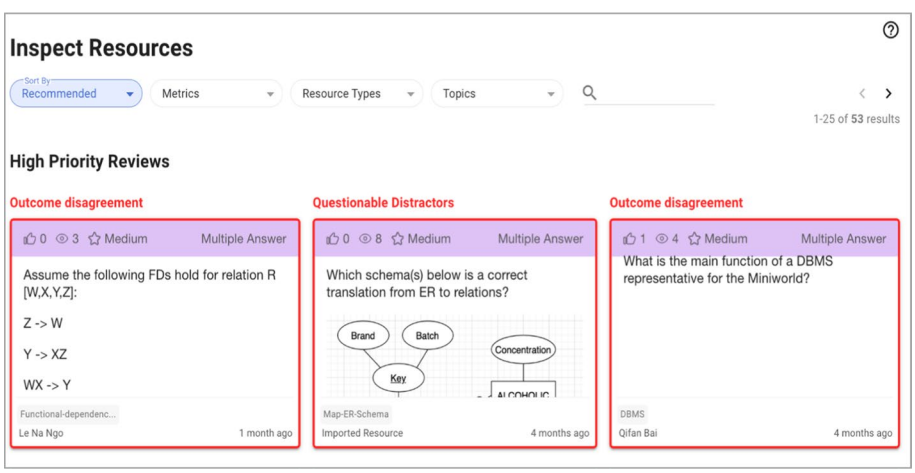

Fig. 7 Interface displaying high-priority resources for review in RiPPLE

review, RiPPLE provides comparisons both in absolute and relative terms, exemplified by statements like, “The level of disagreement in student peer ratings for this resource is 2.8 times the course average". To date, instructors have acted on 11,681 resources flagged for inspection. Of these, 10,463 resources had their outcomes

revised, demonstrating the spot-checking algorithm's 90% effectiveness in identifying resources that require instructor oversight.

Figure 8 showcases RiPPLE's interface for peer feedback exploration, providing instructors with a robust tool to manage and evaluate peer reviews. This interface enables instructors to specify a date range and search for feedback based on word count and specific topics. Furthermore, leveraging the techniques outlined by Darvishi, et al. (2022b), instructors can apply machine learning algorithms to identify reviews that lack constructive suggestions, potentially flagging them as ineffective. Once a set of reviews is selected, instructors have several actionable options: they can up-vote a review to acknowledge its quality, down-vote it while providing feedback on its shortcomings, remove the review entirely—ensuring it has no influence on the final decision and does not count towards the reviewer's assessment requirements—or simply ignore it, thereby excluding it from future searches. To date, instructors have taken action on 19,301 instances of peer feedback, with the most frequent action, occurring in 45% of cases, being the removal of ineffective comments.

Case study summary

The RiPPLE case study demonstrates the effective use of advanced machine learning and AI to refine the peer assessment process within educational frameworks. Table 1 summarizes the various approaches discussed throughout the case study, grouped by different framework areas.

By employing graph-based trust propagation, RiPPLE ensures the reliability of assessors, thus enhancing the credibility and accuracy of peer reviews with diverse and expert insights. The platform enhances individual review processes by integrating generative AI for immediate, constructive feedback, while also allowing for a critical evaluation of this feedback. For assigning grades and feedback, RiPPLE adopts a consensus approach among moderators, relying on a sophisticated trust model to infer reliability and make balanced decisions. Instructor oversight is facilitated through targeted recommendations in the "Suggested Actions" section and a comprehensive peer feedback exploration interface, enabling instructors to effectively manage review quality and student engagement.

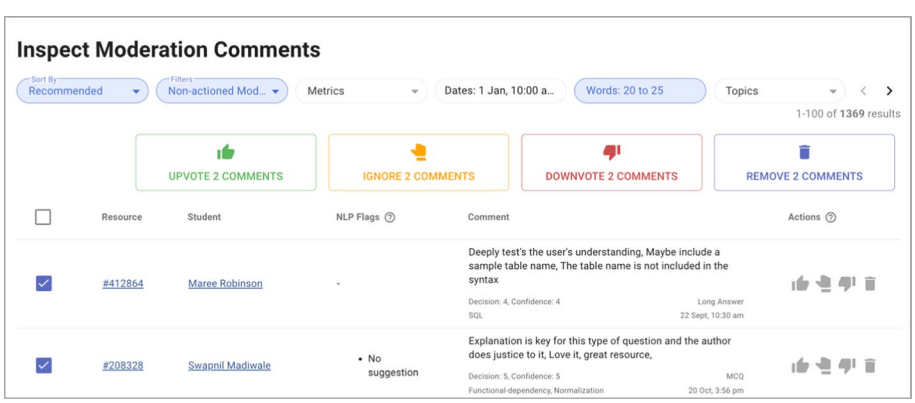

Fig. 8 Interface for the inspection and moderation of peer feedback in RiPPLE

Table 1 Summary of Examples from Ripple in Framework Areas

Framework area	Examples from RiPPLE
Assigning Peer Assessors	Advanced machine learning, including graph-based trust propagation, assesses assessor reliability. This enhances the credibility and accuracy of the peer review process
Enhancing Individual Reviews	RiPPLE integrates generative AI to enhance peer feedback, providing immediate, constructive suggestions for improvement. The AI feedback agent introduces itself, proposes enhancements, warns students of potential inaccuracies, and provides students the ability to provide feedback on its helpfulness
Deriving Peer Grades/Feedback	RiPPLE requires consensus among multiple moderators to finalize resource evaluations. Currently, RiPPLE uses a graph-based trust propagation approach (Darvishi et al., 2021) that infers the reliability of each moderator. The final decision is derived from a weighted average of the ratings given by peer evaluators
Facilitating Instructor Oversight	"Suggested Actions" section offers recommendations for instructors to inspect flagged resources, review ineffective evaluations, identify underperforming students, and remind students of incomplete tasks. The peer feedback exploration interface allows for detailed feedback analysis, offering tools to up-vote quality reviews, down-vote or remove ineffective feedback, and filter reviews for efficiency, with significant instructor engagement in managing peer feedback quality

Table 2 Number of Papers in Areas and Sub-Categories

Area	Number of papers	Sub-category	Number of Sub-category papers
Assigning Peer Assessors	4		
Enhancing Individual Reviews	7		
Deriving Peer Grades/Feedback	35	Automated Assessment	4
		Diversity of Grades and Feedback	7
		Calibration	5
		Fuzzy Logic and Decision-Making	8
		Teamwork Effectiveness	4
		MOOCs	4
		Strategic and Rogue Reporting	3
Analyzing Student Feedback	19	Analysis of Feedback	14
		Automated Feedback	3
		Adaptive Comparative Judgment	2
Facilitating Instructor Oversight	4		
Peer Assessment Systems	10		
Total	79		

Discussion and interpretation of the whole paper

Summary

This paper proposed a theoretical framework for artificial intelligence in peer assessment with six areas: Assigning Peer Assessors, Enhancing Individual Reviews, Deriving Peer Grades/Feedback, Analyzing Student Feedback, Facilitating Instructor Oversight, and Peer Assessment Systems. These six then served as an organizing structure for the following rapid scoping review, within which some areas were further split into sub-categories (Table 2 below summarizes this). A case study followed which further illuminated five of the six areas. The vast majority of the 79 papers in the review found artificial intelligence improved peer assessment. Only two papers

found AI outcomes only as good as or worse than non-AI outcomes. Of course, this might have been expected given publication bias and does not necessarily reflect what practitioners will experience when implementing in their own classrooms.

Within the areas, Deriving Peer Grades/Feedback was by far the most populated. However, most papers focused on dealing with diversity in grades and feedback or with applications of fuzzy logic. Within each area, the main takeaways can be derived from the text. In Diversity of Grades and Feedback, for example, we find that AI can identify weak points in the review and the reviewer, identify problem statements in peer assessments and analyze the nature of peer helpfulness. It can also deal with strategic reporting by comparing peer assessments from more than one reviewer to determine if any assessment is an outlier. AI can weigh the effects of the number of peer assessors, as more reduces reliability but increases validity, irrespective of the burden of effort on students. AI can also explore the assessment of open-ended items to seek higher reliability.

Automated assessment, calibration and team effectiveness seemed under-researched. Similarly in the second largest area Analyzing Student Feedback, the focus was very much on the analysis of feedback with a view to equalizing its quality, while truly automated feedback had very few papers. Improving the Assessment Process and Enhancing Individual Reviews were next largest. Assigning Peer Assessors and Facilitating Instructor Oversight were tied for least populated category. This picture suggests that AI is making progress in peer assessment, but there is still a considerable way to go, particularly with automated assignment and automated assessment. Calibration and teamwork effectiveness also merit further research.

Turning to the case study using the RIPPLE tool, this enables instructors to partner with their students to create pools of high-quality learning resources, each of which are peer assessed and used to recommend personalized content to students based on their mastery level. RIPPLE engages students in a trio of intertwined activities: creation, review and personalized practice. It generates a variety of resource types, including multiple-choice questions, worked examples, flashcards, hotspots and short and long answer questions. The peer assessments are structured around a comprehensive rubric. All activities are conducted under the guidance of instructors, complemented by AI-driven real-time feedback and personalized instructions. The AI feedback component emphasizes that the feedback provided is only a guide and may be inaccurate and asks students to assess whether the feedback is correct.

Strengths and limitations

Our theoretical framework of six areas is of course not the only one possible. Further, despite our attempts to exclude papers only reporting simple digital technology in favor of those reporting artificial intelligence, the boundary was not always clear and subjective interpretation may have played a part. We only searched one database (although we considered this adequate for a scoping review) and used only one coder (albeit with extensive expertise in this area). Sample size in the traditional sense varied considerably, from 55 to over 1,000. However, some studies reported the number of feedback comments (up to 408,000) or peer grades (up to 63,200) analyzed or number of students participating over several years of implementation (up to 12,000). The number of students in MOOCs could be very large. Case studies were also frequent, and while they might be

illuminative, larger quasi-experimental studies are also needed. Samples were mostly of convenience.

Some papers reported only student perceptions, but most had more objective data and some had both. However, many studies only reported on the application of AI to one subject or course, and its utility in other subjects or courses was unknown (although a few studies did explore transfer to other subjects). A majority of studies came from English-speaking Western countries (a consequence of limiting the survey to English-language papers), with China and Taiwan also well represented, but relatively few from other countries or cultures, so the wider applicability to other countries is uncertain. Cultural differences might affect the applicability of AI in other regions. For instance, AI might be better received in China, where traditionally there has been less emphasis on students interrogating the teacher.

Many of these studies compared intervention periods to baseline periods for the same cohort and control groups were present in some cases but a relative rarity. Almost no studies considered the issue of long-term follow-up—did any cognitive or attainment gains endure, especially when AI-fostered peer assessment was no longer operational? Or was there a generalized positive effect from first exposure to AI, in that students might begin experimenting with it successfully of their own volition? We could of course have had tighter inclusion criteria, and only offered a review on a small number of papers, but felt that this would not do justice to the field. The issue of possible publication bias is also of importance, since irrespective of implementation quality, practitioners might not find published research findings reflected in real life.

Opportunities and challenges for researchers and practitioners

For researchers

We have noted above that automated assignment, automated assessment, calibration and teamwork effectiveness merit further research. For example, with respect to automated assessment, in addition to grading and/or giving feedback on the quality of each assessor's feedback in a group setting in order to weight subsequent assessments (e.g., Kalella, 2009; Hoang et al., 2022), could such grades/feedback be fed back to the assessors on a regular basis, in order that they could see their competence increasing and hopefully be motivated to become persistently better assessors? It is important that the opaqueness of AI does not disengage students from active participation in developing their own futures. Regarding methodology, case studies should be accompanied by studies with a more substantial sample size (the median was about $n = 150$), data collected on objective measures as well as perceptions to afford triangulation of data sources, control groups used wherever possible to eliminate the possibility of Hawthorne Effects, follow-up built into the research design to explore transfer and generalization whether positive or negative, transfer across subjects investigated since many studies were subject-specific and transfer across countries investigated given possible cross-cultural differences.

Future studies may need to use multiple methods to assess different aspects of the AI/peer assessment interface. For example, Lin et al. (2024) deployed a four-step approach to reliability detection: recurrent convolutional neural networks (RCNN) were used to detect grade consistency, bidirectional encoder representations from transformers (BERT) were used to detect text originality, long short-term memory (LSTM) was

used to detect grade-text consistency and automated grading was designed with the BERT-RCNN model. It is also worth remembering that generative AI has its dangers. For example, Oviedo-Trespalacios et al. (2023) analyzed ChatGPT's safety-related advice and raised misuse concerns. ChatGPT appeared not to favor contents based on their factuality or reliability. Populations with lower literacy and education were at higher risk of consuming unreliable content.

For practitioners

New and better developed applications appear by the day, which is both exciting for practitioners but also enormously confusing. How can they possibly keep up? Practitioners need to focus on applications which have been demonstrated to have effects by reputable research findings, which probably means that very new applications will not have had time to generate such research. Advertising should be scrutinized with great care. Word-of-mouth recommendations need to be accepted only from colleagues with substantial history of actual implementation. For example, Lin (2022) introduced IdiomsTube (https://www.idiomstube.com), a language learning tool for facilitating the learning of English idiomatic expressions from YouTube videos. This automatically: (1) assesses the speech rate and lexical difficulty level of any learner-chosen English-captioned YouTube video, (2) generates a range of vocabulary-building and revision tasks (e.g., fill-in-the-blanks, the spelling game hangman, pronunciation, flashcards), and (3) recommends YouTube videos based on learners' individual learning progress. Plagiarism detectors for student writing are now being replaced by a new generation of AI detectors which may defeat some of the potentially malign influences of AI (e.g., Rumi, intended to detect the use of tools like ChatGPT in student essay writing – https://www.rumiessay.com), although they may themselves have inherent inaccuracies and problems. We hope that this paper will give practitioners an overview of the rapidly developing field which will help them keep their heads above water.

Conclusion

While AI clearly has advantages, there are also disadvantages, one of which is the lack of transparency. Most AI applications cannot explain how they have reached a decision. Even an expert cannot explain how they produced their outputs. Consequently, their reliability always seems to be in question. As AI develops, its quality in general is likely to improve, but that does not mean it has improved in whatever application is being deployed. Educational institutions typically have a long lead-in time, so many instructors may find themselves using elderly AI. Nonetheless, this paper demonstrates that AI in peer assessment has already achieved a good deal (for example, in Diversity of Grades/Feedback) and has enormous future potential (especially if currently relatively neglected areas like Automated Assignment, Automated Assessment and Calibration become the focus of more research). In general, it seems that AI can produce results as good as real students and instructors, at least in research studies. Whether this will be the case in every practical application is another issue. The goal of peer assessment is to enable formative and/or summative assessment with as much reliability as instructor assessment while developing metacognitive and critical thinking skills in the participants. It is of course a key question as to whether the introduction of AI enhances these goals.

A good deal of research seems more interested in the time-saving for instructor aspects rather than the added value. Much further research work remains to be done.

Acknowledgements
Not applicable.

Author contribution
Topping was the lead writer for the Introduction and Conclusion, Gehringer and his team were principally responsible for the Scoping Review, and Khosravi was the lead writer for the Case Study. All sections were then reviewed by all authors to ensure consensual agreement on the text.

Funding
This research received no external funding.

Availability of data and materials
The articles referred to in this paper are all freely available in the public domain.

Declarations

Competing interests
The authors declare no competing interests.

Received: 2 January 2024 Accepted: 10 October 2024
Published online: 21 January 2025

References

References in the scoping review asterisked *

Abdi, S., Khosravi, H., Sadiq, S., & Demartini, G. (2021). Evaluating the quality of learning resources: A learnersourcing approach. *IEEE Transactions on Learning Technologies, 14*(1), 81–92. https://doi.org/10.1109/TLT.2021.3058644

Abdi, S., Khosravi, H., Sadiq, S., & Gasevic, D. (2020). Complementing educational recommender systems with open learner models. In *Proceedings of the Tenth International Conference on Learning Analytics & Knowledge* (pp. 360–365). https://doi.org/10.1145/3375462.3375520

*Ahuja, R., Khan, D., Symonette, D., Pan, S., Stacey, S., & Engel, D. (2020a). Towards the automatic assessment of student teamwork. In *Companion Proceedings of the 2020 ACM International Conference on Supporting Group Work* (pp. 143–146). https://doi.org/10.1145/3323994.3369894

*Ahuja, R., Khan, D., Tahir, S., Wang, M., Symonette, D., Pan, S., & Engel, D. (2020b). Machine learning and student performance in teams. In: Bittencourt, I., Cukurova, M., Muldner, K., Luckin, R., Millán, E. (Eds.) *Artificial Intelligence in Education*. AIED 2020. Lecture Notes in Computer Science, 12164. Springer, Cham. https://doi.org/10.1007/978-3-030-52240-7_55

*Anaya, A. R., Luque, M., Letón, E., & Hernández-del-Olmo, F. (2019). Automatic assignment of reviewers in an online peer assessment task based on social interactions. *Expert Systems, 36*, e12405. https://doi.org/10.1111/exsy.12405

*Babik, D., Stevens, S. P., Waters, A., & Tinapple, D. (2020). The effects of dispersion and reciprocity on assessment fidelity in peer-review systems: A simulation study. *IEEE Transactions on Learning Technologies, 13*(3), 580–592. https://doi.org/10.1109/TLT.2020.2971495

*Babo, R., Rocha, J., Fitas, R., Suhonen, J., & Tukiainen, M. (2021). Self and peer e-assessment: A study on software usability. *International Journal of Information and Communication Technology Education (IJICTE), 17*(3), 68–85. https://doi.org/10.4018/IJICTE.20210701.oa5

*Badea, G., & Popescu, E. (2020). Supporting students by integrating an open learner model in a peer assessment platform. In: Kumar, V., & Troussas, C. (Eds.) *Intelligent Tutoring Systems*. ITS 2020. Lecture Notes in Computer Science, 12149. Springer, Cham. https://doi.org/10.1007/978-3-030-49663-0_14

*Badea, G., & Popescu, E. (2022). A hybrid approach for mitigating learners' rogue review behavior in peer assessment. In: Crossley, S., & Popescu, E. (Eds.) *Intelligent Tutoring Systems*. ITS 2022. Lecture Notes in Computer Science, 13284. Springer, Cham. https://doi.org/10.1007/978-3-031-09680-8_3

*Bawabe, S., Wilson, L., Zhou, T., Marks, E., & Huang, J. (2021). The UX factor: Using comparative peer review to evaluate designs through user preferences. *Proceedings of the ACM on Human-Computer Interaction, 5* (CSCW2), Article No: 476, 1–23. https://doi.org/10.1145/3479863

*Burrell, N., & Schoenebeck, G. (2021). Measurement integrity in peer prediction: A peer assessment case study. In *Proceedings of the 24th ACM Conference on Economics and Computation* 369–389. https://doi.org/10.1145/3580507.3597744

*Campos, D. G., et al. (2024). Screening smarter, not harder: A comparative analysis of machine learning screening algorithms and heuristic stopping criteria for systematic reviews in educational research. *Educational Psychology Review*. https://doi.org/10.1007/s10648-024-09862-5

*Capuano, N., Loia, V., & Orciuoli, F. (2017). A fuzzy group decision making model for ordinal peer assessment. *IEEE Transactions on Learning Technologies, 10*(2), 247–259. https://doi.org/10.1109/TLT.2016.2565476

*Castro, M. S. O., Mello, R. F., Fiorentino, G., Viberg, O., Spikol, D., Baars, M., & Gašević, D. (2023). Understanding peer feedback contributions using natural language processing. In: Viberg, O., Jivet, I., Muñoz-Merino, P., Perifanou, M.,

Papathoma, T. (Eds.), *Responsive and Sustainable Educational Futures*. EC-TEL 2023. Lecture Notes in Computer Science, vol 14200. Springer, Cham. https://doi.org/10.1007/978-3-031-42682-7_27

*Chai, K. C., & Tay, K. M. (2014). A perceptual computing-based approach for peer assessment. In *9th International Conference on System of Systems Engineering (SOSE)*, Glenelg, SA, Australia, 60–165. https://doi.org/10.1109/SYSOSE.2014.6892481.

*Chai, K. C., Tay, K. M., & Lim, C. P. (2015). A new fuzzy peer assessment methodology for cooperative learning of students. *Applied Soft Computing, 32*, 468–480. https://doi.org/10.1016/j.asoc.2015.03.056

*Chiu, M. M., Woo, C. K., Shiu, A., Liu, Y., & Luo, B. X. (2020). Reducing costly free-rider effects via OASIS. *International Journal of Comparative Education and Development, 22*(1), 30–48. https://doi.org/10.1108/IJCED-07-2019-0041

*Cho, K., & Schunn, C. D. (2007). Scaffolded writing and rewriting in the discipline: A web-based reciprocal peer review system. *Computers & Education, 48*(3), 409–426. https://doi.org/10.1016/j.compedu.2005.02.004

Craig, C. D., & Kay, R. (2021). Examining peer assessment in online learning for higher education - A systematic review of the literature. *Proceedings of ICERI2021 Conference*, 8th-9th November 2021.

Darvishi, A., Khosravi, H., Rahimi, A., Sadiq, S., & Gašević, D. (2022a). Assessing the quality of student-generated content at scale: A comparative analysis of peer-review models. *IEEE Transactions on Learning Technologies, 16*(1), 106–120. https://doi.org/10.1109/TLT.2022.3229022

Darvishi, A., Khosravi, H., & Sadiq, S. (2021). Employing peer review to evaluate the quality of student generated content at scale: A trust propagation approach. In *Proceedings of the Eighth ACM Conference on Learning@ Scale* (pp. 139–150). https://doi.org/10.1145/3491140.3528286

*Darvishi, A., Khosravi, H., Sadiq, S., & Gašević, D. (2022b). Incorporating AI and learning analytics to build trustworthy peer assessment systems. *British Journal of Educational Technology, 53*(4), 844–875. https://doi.org/10.1111/bjet.13233

*Demonacos, C., Ellis, S., & Barber, J. (2019). Student peer assessment using Adaptive Comparative Judgment: Grading accuracy versus quality of feedback. *Practitioner Research in Higher Education, 12*(1), 50–59.

*Dhull, K., Jecmen, S., Kothari, P., & Shah, N. B. (2022). Strategyproofing peer assessment via partitioning: The price in terms of evaluators' expertise. *In Proceedings of the AAAI Conference on Human Computation and Crowdsourcing, 10*(1), 53–63. https://doi.org/10.1609/hcomp.v10i1.21987

*Djelil, F., Brisson, L., Charbey, R., Bothorel, C., Gilliot, J. M., & Ruffieux, P. (2021). Analysing peer assessment interactions and their temporal dynamics using a graphlet-based method. In: De Laet, T., Klemke, R., Alario-Hoyos, C., Hilliger, I., & Ortega-Arranz, A. (Eds.), *Technology-Enhanced Learning for a Free, Safe, and Sustainable World*. EC-TEL 2021. Lecture Notes in Computer Science, 12884. Springer, Cham. https://doi.org/10.1007/978-3-030-86436-1_7

*El Alaoui, M., El Yassini, K., & Ben-Azza, H. (2018). Enhancing MOOCs peer reviews validity and reliability by a fuzzy coherence measure. In *Proceedings of the 3rd International Conference on Smart City Applications, 2018,* Article No.: 57, 1–5. https://doi.org/10.1145/3286606.3286834

*Ellison, C. (2023). Effects of adaptive comparative judgement on student engagement with peer formative feedback. *Practitioner Research in Higher Education, 15*(1), 24–35.

*Fu, Q. K., Lin, C. J., & Hwang, G. J. (2019). Research trends and applications of technology-supported peer assessment: A review of selected journal publications from 2007 to 2016. *Journal of Computers in Education, 6*, 191–213. https://doi.org/10.1007/s40692-019-00131-x

*Garcia-Souto, M. P. (2019). Making assessment of group work fairer and more insightful for students and time-efficient for staff with the new IPAC software. In *INTED2019 Proceedings* (pp. 8636–8641), IATED, Valencia, Spain. https://doi.org/10.21125/inted.2019.2154

*Hamer, J., Kell, C., & Spence, F. (2005). Peer assessment using Aropä. *Ninth Australasian Computing Education Conference (ACE2007), Ballarat, Victoria, Australia, February 2007.* https://www.academia.edu/2878638/Peer_assessment_using_arop%C3%A4?auto=download&email_work_card=download

*He, Y., Hu, X., & Sun, G. (2019). A cognitive diagnosis framework based on peer assessment. In *Proceedings of the ACM Turing Celebration Conference-China*, Article No: 78, 1–6. New York, NY. Association for Computing Machinery. https://doi.org/10.1145/3321408.3322850

Helden, G. V., Van Der Werf, V., Saunders-Smits, G. N., & Specht, M. M. (2023). The use of digital peer assessment in higher education - An umbrella review of literature. *IEEE Access, 11*, 22948–22960. https://doi.org/10.1109/ACCESS.2023.3252914

*Hernández-González, J., & Herrera, P. J. (2023). On the supervision of peer assessment tasks: An efficient instructor guidance technique. in *IEEE Transactions on Learning Technologies*, https://doi.org/10.1109/TLT.2023.3319733.

*Hoang, L. P., Le, H. T., Van Tran, H., Phan, T. C., Vo, D. M., Le, P. A., & Pong-Inwong, C. (2022). Does evaluating peer assessment accuracy and taking it into account in calculating assessor's final score enhance online peer assessment quality? *Education and Information Technologies, 27*, 4007–4035. https://doi.org/10.1007/s10639-021-10763-1

*Hsia, L. H., Huang, I., & Hwang, G. J. (2016). Effects of different online peer-feedback approaches on students' performance skills, motivation and self-efficacy in a dance course. *Computers & Education, 96*, 55–71. https://doi.org/10.1016/j.compedu.2016.02.004

Hua, X., Nikolov, M., Badugu, N., & Wang, L. (2019). Argument mining for understanding peer reviews. arXiv:1903.10104. https://doi.org/10.48550/arXiv.1903.10104

*Huang, C., Tu, Y., Han, Z., Jiang, F., Wu, F., & Jiang, Y. (2023). Examining the relationship between peer feedback classified by deep learning and online learning burnout. *Computers & Education, 207*, 104910. https://doi.org/10.1016/j.compedu.2023.104910

Hwang, G. J., Xie, H., Wah, B. W., & Gašević, D. (2020). Vision, challenges, roles and research issues of Artificial Intelligence in Education. *Computers and Education: Artificial Intelligence, 1*, 100001. https://doi.org/10.1016/j.caeai.2020.100001

*Jónás, T., Tóth, Z. E., & Árva, G. (2018). Applying a fuzzy questionnaire in a peer review process. *Total Quality Management & Business Excellence, 29*(9–10), 1228–1245. https://doi.org/10.1080/14783363.2018.1487616

*Joyner, D. (2018). Intelligent evaluation and feedback in support of a credit-bearing MOOC. In *Artificial Intelligence in Education: 19th International Conference, AIED 2018, London, UK, June 27–30, 2018, Proceedings, Part II 19* (166-170). Springer International Publishing. https://doi.org/10.1007/978-3-319-93846-2_30

*Kalella, T., Lehtonen, T., Luostarinen, P., Riitahuhta, A., & Lanz, M. (2009). Introduction and evaluation of the peer evaluation tool. *New Pedagogy*, 287–292

Khosravi, H., Kitto, K., & Williams, J. J. (2019). RiPPLE: A crowdsourced adaptive platform for recommendation of learning activities. *Journal of Learning Analytics, 6*(3), 91–105. https://doi.org/10.18608/jla.2019.63.12

Khosravi, H., Demartini, G., Sadiq, S., & Gasevic, D. (2021). Charting the design and analytics agenda of learnersourcing systems. *In LAK21: 11th International Learning Analytics and Knowledge Conference,* 32–42. https://doi.org/10.1145/3448139.3448143

Khosravi, H., Denny, P., Moore, S., & Stamper, J. (2023). Learnersourcing in the age of AI: Student, educator and machine partnerships for content creation. *Computers and Education: Artificial Intelligence, 5*, 100151. https://doi.org/10.1016/j.caeai.2023.100151

Kim, S. M., Pantel, P., Chklovski, T., & Pennacchiotti, M. (2006). Automatically assessing review helpfulness. In *Proceedings of the 2006 Conference on Empirical Methods in Natural Language Processing* (pp. 423–430).

*Knight, S., Leigh, A., Davila, Y. C., Martin, L. J., & Krix, D. W. (2019). Calibrating assessment literacy through benchmarking tasks. *Assessment and Evaluation in Higher Education, 44*(8), 1121 – 1132. http://hdl.handle.net/10453/130201

*Kulkarni, C., Wei, K. P., Le, H., Chia, D., Papadopoulos, K., Cheng, J., Koller, D., & Klemmer, S. R. (2013). Peer and self assessment in massive online classes. *ACM Transactions on Computer-Human Interaction., 20*(6), 331–31. https://doi.org/10.1145/2505057

*Kumar, K., Sharma, B., Khan, G. J., Nusair, S., & Raghuwaiya, K. (2020). An exploration on effectiveness of anonymous peer assessment strategy in online formative assessments. In *2020 IEEE Frontiers in Education Conference (FIE),* Uppsala, Sweden. 1–5. IEEE. https://doi.org/10.1109/FIE44824.2020.9274162.

*Lauw, H. W., Lim, E. P., & Wang, K. (2007). Summarizing review scores of "unequal" reviewers. In *Proceedings of the 2007 SIAM International Conference on Data Mining* (pp. 539–544). Society for Industrial and Applied Mathematics. https://doi.org/10.1137/1.9781611972771.58

Li, H. L., Xiong, Y., Hunter, C. V., Xiuyan Guo, X. Y., & Tywoniw, R. (2020a). Does peer assessment promote student learning? A meta-analysis. *Assessment & Evaluation in Higher Education, 45*(2), 193–211. https://doi.org/10.1080/02602938.2019.1620679

*Li, P., Yin, Z., & Li, F. (2020). Quality control method for peer assessment system based on multi-dimensional information. In: Wang, G., Lin, X., Hendler, J., Song, W., Xu, Z., & Liu, G. (Eds.), *Web Information Systems and Applications*. WISA 2020. Lecture Notes in Computer Science, 12432. Springer, Cham. https://doi.org/10.1007/978-3-030-60029-7_17

Lin, P. (2022). Developing an intelligent tool for computer-assisted formulaic language learning from YouTube videos. *ReCALL, 34*(2), 185–200. https://doi.org/10.1017/S0958344021000252

Lin, Z., Yan, H. B., & Zhao, L. (2024). Exploring an effective automated grading model with reliability detection for large-scale online peer assessment. *Journal of Computer Assisted Learning*. https://doi.org/10.1111/jcal.12970

*Liu, C., Doshi, D., Bhargava, M., Shang, R., Cui, J., Xu, D., & Gehringer, E. (2023). Labels are not necessary: Assessing peer-review helpfulness using domain adaptation based on self-training. In *Proceedings of the 18th Workshop on Innovative Use of NLP for Building Educational Applications (BEA 2023)* 173–183. https://doi.org/10.18653/v1/2023.bea-1.15

*Madan, M., & Madan, P. (2015). Fuzzy viva assessment process through perceptual computing. In *2015 Annual IEEE India Conference (INDICON),* New Delhi, India, 1–6. IEEE. https://doi.org/10.1109/INDICON.2015.7443831.

*Masaki , U., Nguyen, D. T., & Ueno, M. (2019). Maximizing accuracy of group peer assessment using item response theory and integer programming. *The Japanese Society for Artificial Intelligence, 33*. https://doi.org/10.11517/pjsai.JSAI2019.0_4H2E503

Mendoza, L. B., Ortega, M. P., Hormaza, J. M., Soto, S. V. (2020). Trends the use of artificial intelligence techniques for peer assessment. *ICEMIS'20: Proceedings of the 6th International Conference on Engineering & MIS* 2020, September 2020. Article No.: 101, Pages 1–7. https://doi.org/10.1145/3410352.3410837

Misiejuk, K., & Wasson, B. (2023). Learning analytics for peer assessment: A scoping review. In: Noroozi, O., & De Wever, B. (Eds.), *The Power of Peer Learning*. Springer, Champaign, IL. https://doi.org/10.1007/978-3-031-29411-2_2

*Nakayama, M., Sciarrone, F., Uto, M., & Temperini, M. (2020). Impact of the number of peers on a mutual assessment as learner's performance in a simulated MOOC environment using the IRT model. *2020 24th International Conference Information Visualisation (IV). Melbourne, Australia, 2020*, 486–490. https://doi.org/10.1109/IV51561.2020.00084

*Ngu, A. H., Shepherd, J., & Magin, D. (1995). Engineering the "Peers" system: The development of a computer-assisted approach to peer assessment. *Research and Development in Higher Education, 18*, 582–587.

*Nguyen, H., Xiong, W., & Litman, D. (2016). Instant feedback for increasing the presence of solutions in peer reviews. In *Proceedings of the 2016 Conference of the North American Chapter of the Association for Computational Linguistics: Demonstrations,* San Diego, California. 6–10.

*Nguyen, H., Xiong, W., & Litman, D. (2017). Iterative design and classroom evaluation of automated formative feedback for improving peer feedback localization. *International Journal of Artificial Intelligence in Education, 27*, 582–622. https://doi.org/10.1007/s40593-016-0136-6

Ocampo, J. C. G., & Panadero, E. (2023). Web-based peer assessment platforms: What educational features influence learning, feedback and social interaction? In: O. Noroozi and B. de Wever (Eds.), *The Power of Peer Learning*. Champaign, IL: Springer. https://doi.org/10.1007/978-3-031-29411-2_8

Oviedo-Trespalacios, O., Peden, A. E., Cole-Hunter, T., Costantini, A., Haghani, M., Rod, J. E., Kelly, S., Torkamaan, H., Tariq, A., Newton, J. D. A., Gallagher, T., Steinert, S., Filtness, A. J., & Reniers, G. (2023). The risks of using ChatGPT to obtain common safety-related information and advice. *Safety Science, 167*, 106244. https://doi.org/10.1016/j.ssci.2023.106244

Ouyang, F., Zheng, L., & Jiao, P. (2022). Artificial intelligence in online higher education: A systematic review of empirical research from 2011 to 2020. *Education and Information Technologies, 27*(6), 7893–7925. https://doi.org/10.1007/s10639-022-10925-9

*Patchan, M. M., Schunn, C. D., & Clark, R. J. (2017). Accountability in peer assessment: Examining the effects of reviewing grades on peer ratings and peer feedback. *Studies in Higher Education, 43*(12), 2263–2278. https://doi.org/10.1080/03075079.2017.1320374

*Patchan, M. M., Schunn, C. D., & Correnti, R. J. (2016). The nature of feedback: How peer feedback features affect students' implementation rate and quality of revisions. *Journal of Educational Psychology, 108*(8), 1098. https://doi.org/10.1037/edu0000103

*Petkovic, D., Okada, K., Sosnick, M., Iyer, A., Zhu, S., Todtenhoefer, R., & Huang, S. (2012). A machine learning approach for assessment and prediction of teamwork effectiveness in software engineering education. In *2012 Frontiers in Education Conference Proceedings,* Seattle, WA. 1–3. IEEE. https://doi.org/10.1109/FIE.2012.6462205.

*Piech, C., Huang, J., Chen, Z., Do, C., Ng, A., & Koller, D. (2013). Tuned models of peer assessment in MOOCs. *arXiv preprint* arXiv:1307.2579. *https://doi.org/10.48550/arXiv.1307.2579*

Purchase, H., & Hamer, J. (2018). Peer-review in practice: Eight years of Aropä. *Assessment & Evaluation in Higher Education, 43*(7), 1146–1165. https://doi.org/10.1080/02602938.2018.1435776

*Ramachandran, L., Gehringer, E. F., & Yadav, R. K. (2017). Automated assessment of the quality of peer reviews using natural language processing techniques. *International Journal of Artificial Intelligence in Education, 27*, 534–581. https://doi.org/10.1007/s40593-016-0132-x

*Rao, D. H., Mangalwede, S. R., & Deshmukh, V. B. (2017). Student performance evaluation model based on scoring rubric tool for network analysis subject using fuzzy logic. In *2017 International Conference on Electrical, Electronics, Communication, Computer, and Optimization Techniques (ICEECCOT),* Mysuru, India (pp. 1–5). IEEE. https://doi.org/10.1109/ICEECCOT.2017.8284623.

*Rashid, M. P., Gehringer, E. F., Young, M., Doshi, D., Jia, Q., & Xiao, Y. (2021). Peer assessment rubric analyzer: An NLP approach to analyzing rubric items for better peer-review. *2021 19th International Conference on Information Technology Based Higher Education and Training (ITHET)*, Sydney, Australia, 2021, 1–9. https://doi.org/10.1109/ITHET50392.2021.9759679.

*Rashid, M. P., Xiao, Y., & Gehringer, E. F. (2022). Going beyond" Good Job": Analyzing helpful feedback from the student's perspective. *Paper presented at the International Conference on Educational Data Mining (EDM)* (15th, Durham, United Kingdom, Jul 24–27, 2022). ERIC Number: ED624053.

*Ravikiran, M. (2020). Systematic review of approaches to improve peer assessment at scale. *arXiv preprint* arXiv:2001.10617. *https://doi.org/10.48550/arXiv.2001.10617*

Raykar, V. C., Yu, S., Zhao, L. H., Valadez, G. H., Florin, C., Bogoni, L., & Moy, L. (2010). Learning from crowds. *Journal of Machine Learning Research, 11*(4), 1297–1322.

*Rico-Juan, J. R., Gallego, A. J., & Calvo-Zaragoza, J. (2019). Automatic detection of inconsistencies between numerical scores and textual feedback in peer-assessment processes with machine learning. *Computers & Education, 140*, 103609. https://doi.org/10.1016/j.compedu.2019.103609

*Russell, A. R. (2013). The evolution of Calibrated Peer Review. *Trajectories of Chemistry Education Innovation and Reform*, Chapter 9, pp 129–143. American Chemical Society Symposium Series, Vol. 1145. https://doi.org/10.1021/bk-2013-1145.ch009

*Saarinen, S., Krishnamurthi, S., Fisler, K., & Tunnell Wilson, P. (2019). Harnessing the wisdom of the classes: Classsourcing and machine learning for assessment instrument generation. *In Proceedings of the 50th ACM Technical Symposium on Computer Science Education*, 606–612. https://doi.org/10.1145/3287324.3287504

*Saccardi, I., Veth, D., & Masthoff, J. (2023). Identifying students' group work problems: Design and field studies of a supportive peer assessment. *Interacting with Computers*. https://doi.org/10.1093/iwc/iwad044

*Sciarrone, F., & Temperini, M. (2020). A web-based system to support teaching analytics in a MOOC's simulation environment. In *2020 24th International Conference Information Visualisation (IV),* Melbourne, Australia (491–495). IEEE. https://doi.org/10.1109/IV51561.2020.00085.

*Selmi, M., Hage, H., & Aïmeur, E. (2014). Opinion Mining for predicting peer affective feedback helpfulness. In *International Conference on Knowledge Management and Information Sharing,* 2, 419–425. SCITEPRESS. https://doi.org/10.5220/0005158704190425

*Sharma, D., & Potey, M. (2018). Effective learning through peer assessment using Peergrade tool. In *2018 IEEE Tenth International Conference on Technology for Education (T4E),* Chennai, India, 114–117. IEEE. https://doi.org/10.1109/T4E.2018.00031.

*Shishavan, H. B., & Jalili, M. (2020). Responding to student feedback: Individualising teamwork scores based on peer assessment. *International Journal of Educational Research Open, 1*, 100019. https://doi.org/10.1016/j.ijedro.2020.100019

*Siemens, G., Marmolejo-Ramos, F., Gabriel, F., Medeiros, K., Marrone, R., Joksimovic, S., & de Laat, M. (2022). Human and artificial cognition. *Computers and Education: Artificial Intelligence, 3*, 100107. https://doi.org/10.1016/j.caeai.2022.100107

*Stelmakh, I., Shah, N. B., & Singh, A. (2021). Catch me if I can: Detecting strategic behaviour in peer assessment. *In Proceedings of the AAAI Conference on Artificial Intelligence, 35*(6), 4794–4802. https://doi.org/10.1609/aaai.v35i6.16611

*Thamizhkkanal, M. R., & Ambeth Kumar, V. D. (2020). A neural based approach to evaluate an answer script. In: Hemanth, D., Kumar, V., Malathi, S., Castillo, O., & Patrut, B. (Eds.) *Emerging Trends in Computing and Expert Technology*. COMET 2019. Lecture Notes on Data Engineering and Communications Technologies, vol 35. Springer, Cham. https://doi.org/10.1007/978-3-030-32150-5_122

*Tiew, H. B., Chua, F. F., & Chan, G. Y. (2021). G-PAT: A group peer assessment tool to support group projects. In *2021 7th International Conference on Research and Innovation in Information Systems (ICRIIS)*, Johor Bahru, Malaysia, 1–6. https://doi.org/10.1109/ICRIIS53035.2021.9617037.

Topping, K. J. (2023). Peer- and self-assessment. In: volume 9, Assessment and Accountability, Ed. T. Hopfenbeck. (pp. 79–88). In: Tierney, R. J., Rizvi, F., & K. Erkican, K. (Eds.), *International Encyclopaedia of Education* (fourth edition, volume 13). Oxford: Elsevier. https://doi.org/10.1016/B978-0-12-818630-5.09021-7

Topping, K. J. (2024). *Improving thinking about thinking in the classroom: What works for enhancing metacognition*. Routledge.

*Wang, A. Y., Chen, Y., Chung, J. J. Y., Brooks, C., & Oney, S. (2021). PuzzleMe: Leveraging peer assessment for in-class programming exercises. *Proceedings of the ACM on Human-Computer Interaction,* 5, Issue CSCW2, Article No: 415, 1–24. https://doi.org/10.1145/3479559

*Wang, Y., Li, H., Feng, Y., Jiang, Y., & Liu, Y. (2012). Assessment of programming language learning based on peer code review model: Implementation and experience report. *Computers & Education, 59*(2), 412–422. https://doi.org/10.1016/j.compedu.2012.01.007

*Wang, Y. Q., Liu, B. Y., Zhang, K., Jiang, Y. S., & Sun, F. Q. (2019a). Reviewer assignment strategy of peer assessment: Towards managing collusion in self-assignment. *Proceedings of the 2nd International Conference on Social Science, Public Health and Education* (SSPHE 2018). https://doi.org/10.2991/ssphe-18.2019.75

*Wang, R., Wei, S., Ohland, M. W., & Ferguson, D. M. (2019b). Natural language processing system for self-reflection and peer-evaluation. *In the Fourth North American International Conference on Industrial Engineering and Operations Management*, Toronto, Canada, October 23–25, 2019 (pp. 229–238).

*Wei, S., Wang, R., Ohland, M. W., & Nanda, G. (2020). Automating anonymous processing of peer evaluation comments. In *2020 ASEE Virtual Annual Conference Content Access*. https://doi.org/10.18260/1-2--35615

*Wu, C., Chanda, E., & Willison, J. (2010). SPARKPlus for self-and peer assessment on group-based honours' research projects. *The Education Research Group of Adelaide (ERGA) conference 2010: The Changing Face of Education,* 24–25 September, 2010. https://hdl.handle.net/2440/61612

*Xiao, Y., Y., Gao, Y., Yue, C. H., & Gehringer, E. (2022). Estimating student grades through peer assessment as a crowdsourcing calibration problem. *20th International Conference on Information Technology Based Higher Education and Training (ITHET)*, Antalya, Turkey, 2022, pp. 1–9, https://doi.org/10.1109/ITHET56107.2022.10031993.

*Xiao, Y., Zingle, G., Jia, Q., Akbar, S., Song, Y., Dong, M., & Gehringer, E. (2020a). Problem detection in peer assessments between subjects by effective transfer learning and active learning. *The International Conference on Educational Data Mining (EDM)* (13th, Online, Jul 10–13, 2020). ERIC Number: ED608055

*Xiao, Y., Zingle, G., Jia, Q., Shah, H. R., Zhang, Y., Li, T., & Gehringer, E. F. (2020b). Detecting problem statements in peer assessments. *arXiv preprint* arXiv:2006.04532. *https://doi.org/10.48550/arXiv.2006.04532*

Xiong, W., & Litman, D. (2010). Identifying problem localization in peer-review feedback. In V. Aleven, J. Kay, & J. Mostow (Eds.), *Intelligent tutoring systems. ITS 2010. Lecture notes in computer science, 6095*. Berlin: Springer. https://doi.org/10.1007/978-3-642-13437-1_93

*Xiong, W., & Litman, D. (2011). Automatically predicting peer-review helpfulness. In *Proceedings of the 49th Annual Meeting of the Association for Computational Linguistics: Human Language Technologies,* Portland, Oregon, 502–507.

*Xiong, W., Litman, D., & Schunn, C. (2012). Natural language processing techniques for researching and improving peer feedback. *Journal of Writing Research, 4*(2), 155–176. https://doi.org/10.17239/jowr-2012.04.02.3

*Xiong, Y., Schunn, C. D., & Wu, Y. (2023). What predicts variation in reliability and validity of online peer assessment? A large-scale cross-context study. *Journal of Computer Assisted Learning, 39*(6), 2004–2024. https://doi.org/10.1111/jcal.12861

Zheng, L. Q., Zhang, X., & Cui, P. P. (2020). The role of technology-facilitated peer assessment and supporting strategies: A meta-analysis. *Assessment & Evaluation in Higher Education, 45*(3), 372–386. https://doi.org/10.1080/02602938.2019.1644603

*Zingle, G., Radhakrishnan, B., Xiao, Y., Gehringer, E., Xiao, Z., Pramudianto, F., Arnav, A. (2019). Detecting suggestions in peer assessments. *International Conference on Educational Data Mining (EDM)* (12th, Montreal, Canada, Jul 2–5, 2019). ERIC Number: ED599201.

*Zong, Z., & Schunn, C. D. (2023). Does matching peers at finer-grained levels of prior performance enhance gains in task performance from peer review? *International Journal of Computer-Supported Collaborative Learning, 18*, 425–456. https://doi.org/10.1007/s11412-023-09401-4

Publisher's Note

Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.